17. DMA Controller (DMAC)

17.1 Overview

The 8-channel direct memory access controller (DMAC) that can transfer data without intervention from the CPU. When a DMA transfer request is generated, the DMAC transfers data stored at the transfer source address to the transfer destination address.

Table 17.1 lists the DMAC specifications, and Figure 17.1 shows a block diagram of the DMAC.

Table 17.1 DMAC specifications (1 of 2)

<table>
<tr><th colspan="2">Item</th><th>DMAC0 (for CPU0)</th><th>DMAC1 (for CPU1)</th></tr>
<tr><td colspan="2">Number of channels</td><td>8 channels (DMAC0n (n = 0 to 7))</td><td>8 channels (DMAC1n (n = 0 to 7))</td></tr>
<tr><td colspan="2">Transfer space</td><td colspan="2">4 GB
(0x00000000 to 0xFFFFFFFF excluding reserved areas)</td></tr>
<tr><td colspan="2">Maximum transfer volume</td><td colspan="2">64 M data
(Maximum number of transfers in block transfer mode: 1,024 data × 65,536 blocks)</td></tr>
<tr><td colspan="2">DMAC activation source</td><td colspan="2">Selectable for each channel:
● Software trigger
● Interrupt requests from peripheral modules or trigger input from external interrupt input pins.*1</td></tr>
<tr><td colspan="2">Channel priority</td><td colspan="2">Selectable from fixed priority or round-robin
● channel 0 > channel 1 > channel 2 > channel 3 … > channel 7 > DTC (channel 0: Highest)
● Round-robin</td></tr>
<tr><td rowspan="2">Transfer data</td><td>Single data</td><td colspan="2">Bit length: 8, 16, 32, 64 bits</td></tr>
<tr><td>Block size</td><td colspan="2">Number of data: 1 to 1,024</td></tr>
<tr><td rowspan="4">Transfer mode</td><td>Normal transfer mode</td><td colspan="2">● One data transfer by one DMA transfer request
● Free running function (setting in which total number of data transfers is not specified) settable</td></tr>
<tr><td>Repeat transfer mode</td><td colspan="2">● One data transfer by one DMA transfer request
● Program returns to the transfer start address on completion of the repeat size of data transfer specified for the transfer source or destination.
● Maximum settable repeat size: 1,024
● Selectable free running function</td></tr>
<tr><td>Repeat-block transfer mode</td><td colspan="2">● One block data transfer by one DMA transfer request
● Maximum settable block size: 1,024
● Block transfer can be repeated
● Maximum settable repeat size: 64K
● Selectable free running function</td></tr>
<tr><td>Block transfer mode</td><td colspan="2">● One block data transfer by one DMA transfer request
● Maximum settable block size: 1,024 data
● Selectable free running function</td></tr>
<tr><td>Selective functions</td><td>Extended repeat area function</td><td colspan="2">● Function in which data can be transferred by repeating the address values in the specified range with the upper bit values in the transfer address register fixed
● Area of 2 bytes to 128 MB separately settable as extended repeat area for transfer source and destination</td></tr>
<tr><td colspan="2">Processing on DMAC transfer error</td><td colspan="2">● When DMAC transfer error occurs, the channel on which the error occurred is stopped.
● DMAC Event link setting register (DELSRn (n =0 to 7)) in the ICU is cleared. Clearing targets are selectable for each channel or all channels.</td></tr>
<tr><td rowspan="2">Interrupt DMACmn_INT (m = 0, 1, n = 0 to 7)</td><td>Transfer end interrupt</td><td colspan="2">Generated on completion of transferring data volume specified by the transfer counter.</td></tr>
<tr><td>Transfer escape end interrupt</td><td colspan="2">● Generated when the repeat size of data transfer is completed.
● Generated when the source address extended repeat area overflows.
● Generated when the destination address extended repeat area overflows.</td></tr>
<tr><td>Interrupt DMAm_TRANSERR (m = 0, 1)</td><td>Error response detection interrupt</td><td colspan="2">● Generated when the DMAC transfer error occurs</td></tr>
</table>

Table 17.1 DMAC specifications (2 of 2)

Item	DMAC0 (for CPU0)	DMAC1 (for CPU1)
Event link activation (DMACmn_INT (m = 0, 1, n = 0 to 7))	Generated after each data transfer (for block transfer, after each block is transferred).	
Module-stop function	Module-stop state can be set to reduce power consumption.	
TrustZone Filter	Security and Privilege can be set for each channel.	

Note: Security attribution Register of DMAC channel is described in ICU.ICUSARC

Note 1. For details on DMAC activation sources, see Table 14.5 in section 14, Interrupt Controller Unit (ICU).

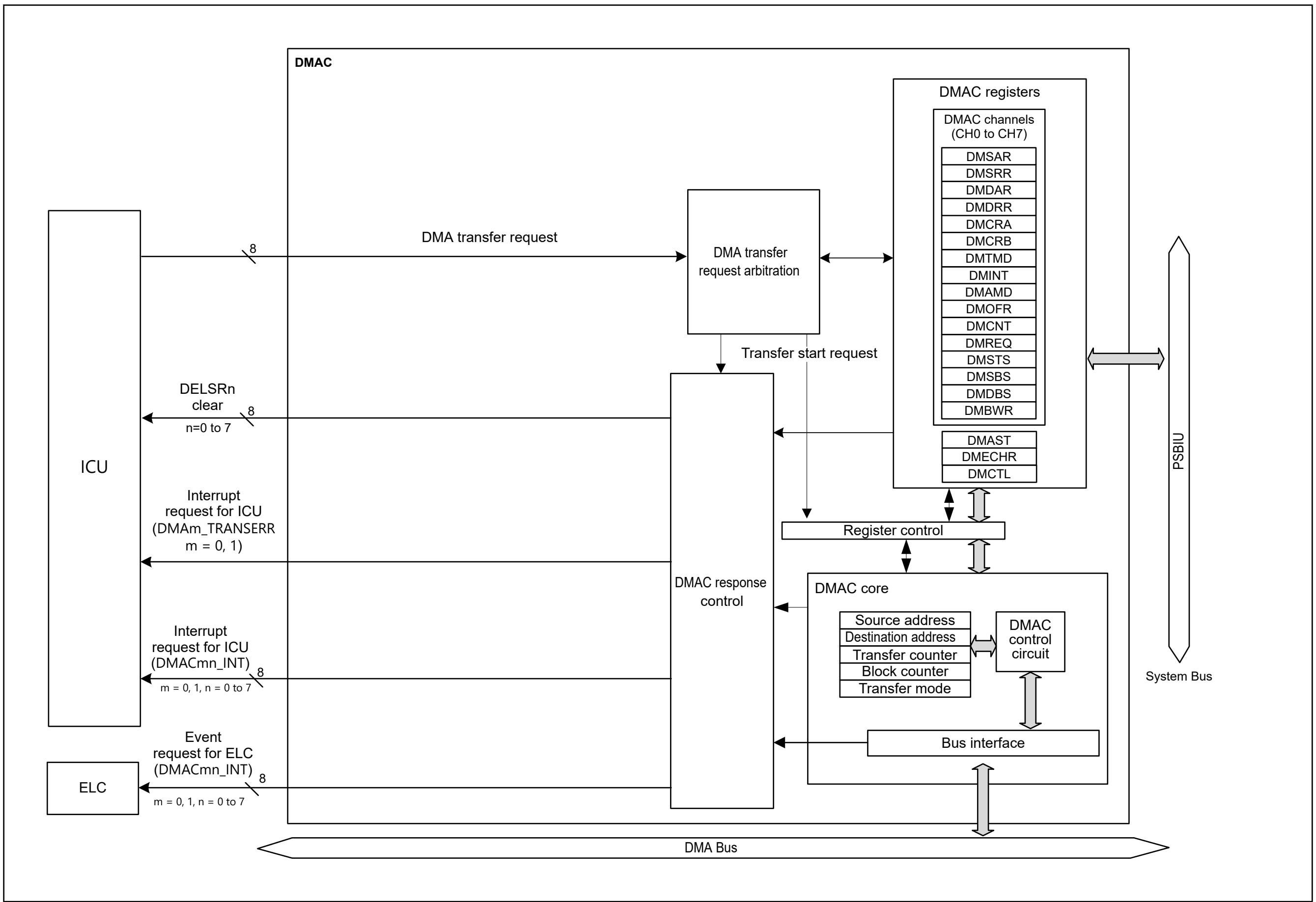

Figure 17.1 DMAC block diagram

17.2 Register Descriptions

The DMAC0 and DMAC1 registers have the same base address. DMAC0 is dedicated to CPU0 and can only be accessed from CPU0. DMAC1 is dedicated to CPU1 and can only be accessed from CPU1.

17.2.1 DMACCHSAR : DMA Channel Security Attribution Register

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x1A0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	SADMAC107	SADMAC106	SADMAC105	SADMAC104	SADMAC103	SADMAC102	SADMAC101	SADMAC100
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	SADMAC007	SADMAC006	SADMAC005	SADMAC004	SADMAC003	SADMAC002	SADMAC001	SADMAC000
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	SADMAC007 to SADMAC000	Security attributes of output and registers for DMAC0 channel 0: Secure 1: Non-secure	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
23:16	SADMAC107 to SADMAC100	Security attributes of output and registers for DMAC1 channel 0: Secure 1: Non-secure	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1

SADMAC00n (n = 0 to 7)

Security attributes registers for DMAC0 channel.

This bit determines the security attribute of the output as the master for each DMAC0 channel.

The controlled channel of DMAC0 registers are shown below. For details, see section 17.3.13. Channel Security.

- CPSCU.DMACCHPAR.PADMAC00n
- ICU0.DELSR channel n
- DMAC0.DMSAR channel n
- DMAC0.DMSRR channel n
- DMAC0.DMDAR channel n
- DMAC0.DMDRR channel n
- DMAC0.DMCRA channel n
- DMAC0.DMCRB channel n
- DMAC0.DMTMD channel n
- DMAC0.DMINT channel n
- DMAC0.DMAMD channel n
- DMAC0.DMOFR channel n
- DMAC0.DMCNT channel n
- DMAC0.DMREQ channel n
- DMAC0.DMSTS channel n
- DMAC0.DMSBS channel n
- DMAC0.DMDBS channel n
- DMAC0.DMBWR channel n

SADMAC10n (n = 0 to 7)

Security attributes registers for DMAC1 channel.

This bit determines the security attribute of the output as the master for each DMAC1 channel.

The controlled channel of DMAC1 registers are shown below. For details, see section 17.3.13. Channel Security.

- CPSCU.DMACCHPAR.PADMAC10n
- ICU1.DELSR channel n
- DMAC1.DMSAR channel n
- DMAC1.DMSRR channel n
- DMAC1.DMDAR channel n
- DMAC1.DMDRR channel n
- DMAC1.DMCRA channel n
- DMAC1.DMCRB channel n
- DMAC1.DMTMD channel n
- DMAC1.DMINT channel n
- DMAC1.DMAMD channel n
- DMAC1.DMOFR channel n
- DMAC1.DMCNT channel n
- DMAC1.DMREQ channel n
- DMAC1.DMSTS channel n
- DMAC1.DMSBS channel n
- DMAC1.DMDBS channel n
- DMAC1.DMBWR channel n

17.2.2 DMACCHPAR : DMA Channel Privilege Attribution Register

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x1F0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	PADMAC107	PADMAC106	PADMAC105	PADMAC104	PADMAC103	PADMAC102	PADMAC101	PADMAC100
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	PADMAC007	PADMAC006	PADMAC005	PADMAC004	PADMAC003	PADMAC002	PADMAC001	PADMAC000
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
7:0	PADMAC007 to PADMAC000	Privilege attributes of outputs and registers for DMAC0 channel 0: Privileged. 1: Unprivileged.	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
23:16	PADMAC107 to PADMAC100	Privilege attributes of outputs and registers for DMAC1 channel 0: Privileged. 1: Unprivileged.	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-2, P-TYPE-1

PADMAC00n (n = 0 to 7)

Privilege attributes of output and registers for DMAC0 channel.

This bit determines the Privilege attribute of the output as the master for each DMAC0 channel. It also determines the privileged attributes of DMAC0 registers for each channel. For details, see section 17.3.14. Channel Privilege.

The controlled the channel of DMAC0 registers are shown below.

- DMAC0.DMSAR channel n
- DMAC0.DMSRR channel n
- DMAC0.DMDAR channel n
- DMAC0.DMDRR channel n
- DMAC0.DMCRA channel n
- DMAC0.DMCRB channel n
- DMAC0.DMTMD channel n
- DMAC0.DMINT channel n
- DMAC0.DMAMD channel n
- DMAC0.DMOFR channel n
- DMAC0.DMCNT channel n
- DMAC0.DMREQ channel n
- DMAC0.DMSTS channel n
- DMAC0.DMSBS channel n
- DMAC0.DMDBS channel n
- DMAC0.DMBWR channel n

PADMAC10n (n = 0 to 7)

Privilege attributes of output and registers for DMAC1 channel.

This bit determines the Privilege attribute of the output as the master for each DMAC1 channel. It also determines the privileged attributes of DMAC1 registers for each channel. For details, see section 17.3.14. Channel Privilege.

The controlled the channel of DMAC1 registers are shown below.

- DMAC1.DMSAR channel n
- DMAC1.DMSRR channel n
- DMAC1.DMDAR channel n
- DMAC1.DMDRR channel n
- DMAC1.DMCRA channel n
- DMAC1.DMCRB channel n
- DMAC1.DMTMD channel n
- DMAC1.DMINT channel n
- DMAC1.DMAMD channel n
- DMAC1.DMOFR channel n
- DMAC1.DMCNT channel n
- DMAC1.DMREQ channel n
- DMAC1.DMSTS channel n
- DMAC1.DMSBS channel n
- DMAC1.DMDBS channel n
- DMAC1.DMBWR channel n

17.2.3 DMACSAR : DMAC Controller Security Attribution Register

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x34

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DMASTSA1
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DMASTSA0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DMASTSA0	DMAC0 DMAST Security Attribution 0: Secure 1: Non-secure	R/W
15:1	—	These bits are read as 0. The write value should be 0.	R/W
16	DMASTSA1	DMAC1 DMAST Security Attribution 0: Secure 1: Non-secure	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1

For DMAC, security attribution is set for each channel. However, this register only sets the DMAST register security attribution.

DMASTSA0 bit (DMAC0 DMAST Security Attribution)

This bit specifies security attribution of the registers for DMAC0.DMAST and DMAC0.DMCTL. Do not write to the DMASTSA0 bit while DMAC0 transfer is enabled or a bus master is writing to the DMAC0 registers.

DMASTSA1 bit (DMAC1 DMAST Security Attribution)

This bit specifies security attribution of the registers for DMAC1.DMAST and DMAC1.DMCTL. Do not write to the DMASTSA1 bit while DMAC1 transfer is enabled or a bus master is writing to the DMAC1 registers.

17.2.4 DMSAR : DMA Source Address Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x00

Bit position: 31 … 0

Bit field: (32-bit field)

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	n/a	Specifies the transfer source start address Setting range is 0x00000000 to 0xFFFFFFFF (4 GB).	R/W

Note: S-TYPE-3, P-TYPE-3

Set DMSAR while DMAC activation is disabled (DMAST.DMST = 0) or DMA transfer is disabled (DMCNT.DTE = 0).

Note: Address alignment in this register must match the Transfer Data Size value selected in the DMTMD.SZ bit.

17.2.5 DMSRR : DMA Source Reload Address Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x20

Bit position: 31 0

Bit field:

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	n/a	Specifies the transfer source reload address 0x0000 0000 to 0xFFFF FFFF (4 GB)	R/W

Note: S-TYPE-3, P-TYPE-3

Set DMSRR while DMAC activation is disabled (DMAST.DMST = 0) or DMA transfer is disabled (DMCNT.DTE = 0).

DMSRR is used to store the start address of the buffer size set in DMSBS during repeat-blcok transfer mode. In repeat-block transfer mode, DMSAR reloads value of DMSRR after specified transfer finished.

In normal transfer mode, repeat transfer mode and block transfer mode, DMSRR is not used. The setting is invalid.

Note: Address alignment in this register must match the Transfer Data Size value selected in the DMTMD.SZ bit.

17.2.6 DMDAR : DMA Destination Address Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x04

Bit position: 31 0

Bit field:

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	n/a	Specifies the transfer destination start address Setting range is 0x0000_0000 to 0xFFFF_FFFF (4 GB).	R/W

Note: S-TYPE-3, P-TYPE-3

Set DMDAR while DMAC activation is disabled (DMAST.DMST = 0) or DMA transfer is disabled (DMCNT.DTE = 0).

Note: Address alignment in this register must match the Transfer Data Size value selected in the DMTMD.SZ bits.

17.2.7 DMDRR : DMA Destination Reload Address Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x24

Bit position: 31 0

Bit field:

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	n/a	Specifies the transfer destination reload address Setting range is 0x0000_0000 to 0xFFFF_FFFF (4 GB).	R/W

Note: S-TYPE-3, P-TYPE-3

Set DMDRR while DMAC activation is disabled (DMAST.DMST = 0) or DMA transfer is disabled (DMCNT.DTE = 0).

DMDRR is used to store the start address of the buffer size set in DMDBS during repeat-blcok transfer mode. In repeat-block transfer mode, DMDAR reloads the value of DMDRR after the specified transfer is finished.

In normal transfer mode, repeat transfer mode and block transfer mode, DMDRR is not used. The setting is invalid.

Note: Address alignment in this register must match the Transfer Data Size value selected in the DMTMD.SZ bits.

17.2.8 DMCRA : DMA Transfer Count Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x08

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	DMCRAH[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DMCRAL[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	DMCRAL[15:0]	Lower bits of transfer count Specifies the number of transfer operations.	R/W
25:16	DMCRAH[9:0]	Upper bits of transfer count Specifies the number of transfer operations.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Set DMCRA while DMAC activation is disabled (DMAST.DMST = 0) or DMA transfer is disabled (DMCNT.DTE = 0).

Set the same value for DMCRAH and DMCRAL in repeat transfer mode, block transfer mode, and repeat-block transfer mode. Bits 15 to 10 are fixed to 0 in repeat transfer mode, block transfer mode, and repeat-block transfer mode.

(1) Normal Transfer Mode (DMTMD.MD[1:0] = 00b)

DMCRAL functions as a 16-bit transfer counter.

The number of transfer operations is one when the setting is 0x0001, and 65,535 when it is 0xFFFF. The value is decremented by one each time data is transferred.

When the setting is 0x0000, no specific number of transfer operations is set; data transfer is performed with the transfer counter stopped (free running function).

Free running function is not selected by DMTMD.TKP bit in normal transfer mode.

DMCRAH is not used in normal transfer mode. Write 0x0000 to DMCRAH.

(2) Repeat Transfer Mode (DMTMD.MD[1:0] = 01b)

DMCRAH specifies the repeat size and DMCRAL functions as a 10-bit transfer counter.

The number of transfer operations is one when the setting is 0x001, 1023 when it is 0x3FF, and 1024 when it is 0x000. In repeat transfer mode, a value in the range of 0x000 to 0x3FF (1 to 1024) can be set for DMCRAH and DMCRAL.

Setting bits 15 to 10 in DMCRAL is invalid. Write 0 to these bits.

The value in DMCRAL is decremented by one each time data is transferred until it reaches 0x000, at which the value in DMCRAH is loaded into DMCRAL.

(3) Block Transfer Mode (DMTMD.MD[1:0] = 10b)

DMCRAH specifies the block size and DMCRAL functions as a 10-bit block size counter.

The block size is one when the setting is 0x001, 1023 when it is 0x3FF, and 1024 when it is 0x000. In block transfer mode, a value in the range of 0x000 to 0x3FF can be set for DMCRAH and DMCRAL.

Setting bits 15 to 10 in DMCRAL is invalid. Write 0 to these bits.

The value in DMCRAL is decremented by one each time data is transferred until it reaches 0x000, at which the value in DMCRAH is loaded into DMCRAL.

(4) Repeat-Block Transfer Mode (DMTMD.MD[1:0] = 11b)

DMCRAH specifies the block size and DMCRAL functions as a 10-bit block size counter.

The block size is one when the setting is 0x001, 1023 when it is 0x3FF, and 1024 when it is 0x000. In repeat-block transfer mode, a value in the range of 0x000 to 0x3FF can be set for DMCRAH and DMCRAL.

Setting bits 15 to 10 in DMCRAL is invalid. Write 0 to these bits.

The value in DMCRAL is decremented by one each time data is transferred until it reaches 0x000, at which the value in DMCRAH is loaded into DMCRAL.

17.2.9 DMCRB : DMA Block Transfer Count Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x0C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	DMCRBH[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DMCRBL[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	DMCRBL[15:0]	Functions as a number of block, repeat or repeat-block transfer counter. 0x0001 to 0xFFFF (1 to 65535) 0x0000 (65536)	R/W
31:16	DMCRBH[15:0]	Specifies the number of block, repeat or repeat-block transfer operations. 0x0001 to 0xFFFF (1 to 65535) 0x0000 (65536)	R/W

Note: S-TYPE-3, P-TYPE-3

Set DMCRB while DMAC activation is disabled (DMAST.DMST = 0) or DMA transfer is disabled (DMCNT.DTE = 0).

Set the same value for DMCRBH and DMCRBL in repeat transfer mode, block transfer mode and repeat-block transfer mode.

DMCRBH specifies the number of block, repeat and repeat-block transfer operations, and DMCRBL functions as a 16-bit the number of block counter in block, repeat, and repeat-block transfer mode, respectively.

The number of transfer operations is one when the setting is 0x0001, 65535 when it is 0xFFFF, and 65536 when it is 0x0000.

In repeat transfer mode, the value is decremented by one when the final data of one repeat size is transferred.

In block transfer mode and repeat-block transfer mode, the value is decremented by one when the final data of one block size is transferred.

In normal transfer mode, DMCRB is not used. The setting is invalid.

When DMTMD.TKP is 1 and the final data of one repeat size or one block size is transferred, DMCRBL reloads the value of DMCRBH automatically.

17.2.10 DMTMD : DMA Transfer Mode Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x10

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MD[1:0]		DTS[1:0]		—	TKP	SZ[1:0]		—	—	—	—	—	—	DCTG[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	DCTG[1:0]	Transfer Request Source Select 0 0: Software request 0 1: Hardware request*1 1 0: Setting prohibited 1 1: Setting prohibited	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
9:8	SZ[1:0]	Transfer Data Size Select 0 0: 8 bits 0 1: 16 bits 1 0: 32 bits 1 1: 64 bits	R/W
10	TKP	Transfer Keeping 0: Transfer is stopped by completion of specified total number of transfer operations. 1: Transfer is not stopped by completion of specified total number of transfer operations (free-running).	R/W
11	—	This bit is read as 0. The write value should be 0.	R/W
13:12	DTS[1:0]	Repeat Area Select 0 0: The destination is specified as the repeat area or block area. 0 1: The source is specified as the repeat area or block area. 1 0: The repeat area or block area is not specified. 1 1: Setting prohibited.	R/W
15:14	MD[1:0]	Transfer Mode Select 0 0: Normal transfer 0 1: Repeat transfer 1 0: Block transfer 1 1: Repeat-block transfer	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. To select the DMAC activation source, use the ICU.DELSRn registers. For details on DMAC activation sources, see Table 14.5 in section 14, Interrupt Controller Unit (ICU).

Set DMTMD while DMAC activation is disabled (DMAST.DMST = 0) or DMA transfer is disabled (DMCNT.DTE = 0).

DTS[1:0] bits (Repeat Area Select)

DTS[1:0] select either the source or destination as the repeat area in repeat or block transfer mode. In normal or repeat-block transfer mode, setting these bits is invalid.

TKP bit (Transfer Keeping)

TKP selects either stopping transfer or keeping transfer by completion of specified total number of transfer operations in repeat, block or repeat-block transfer mode. In normal transfer mode, setting this bit is invalid.

17.2.11 DMINT : DMA Interrupt Setting Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x13

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	DTIE	ESIE	RPTIE	SARIE	DARIE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DARIE	Destination Address Extended Repeat Area Overflow Interrupt Enable 0: Disables an interrupt request for an extended repeat area overflow on the destination address. 1: Enables an interrupt request for an extended repeat area overflow on the destination address.	R/W
1	SARIE	Source Address Extended Repeat Area Overflow Interrupt Enable 0: Disables an interrupt request for an extended repeat area overflow on the source address. 1: Enables an interrupt request for an extended repeat area overflow on the source address.	R/W
2	RPTIE	Repeat Size End Interrupt Enable 0: Disables the repeat size end interrupt request. 1: Enables the repeat size end interrupt request.	R/W
3	ESIE	Transfer Escape End Interrupt Enable 0: Disables the transfer escape end interrupt request. 1: Enables the transfer escape end interrupt request.	R/W
4	DTIE	Transfer End Interrupt Enable 0: Disables the transfer end interrupt request. 1: Enables the transfer end interrupt request.	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Set DMINT while DMAC activation is disabled (DMAST.DMST = 0) or DMA transfer is disabled (DMCNT.DTE = 0).

DARIE bit (Destination Address Extended Repeat Area Overflow Interrupt Enable)

When an extended repeat area overflow on the destination address occurs while DARIE bit is set to 1, the DMCNT.DTE bit is cleared to 0. At the same time, the DMSTS.ESIF flag is set to 1 to indicate that an interrupt by an extended repeat area overflow on the destination address is requested.

When block transfer mode is used with the extended repeat area function, an interrupt is requested after completion of a 1-block size transfer. When setting 1 in the DMCNT.DTE bit of the channel for which a transfer has been stopped, the transfer is resumed from the state when the transfer is stopped.

When the extended repeat area is not specified for the destination address, this bit is ignored.

Do not set this bit to 1 in repeat-block transfer mode.

SARIE bit (Source Address Extended Repeat Area Overflow Interrupt Enable)

When an extended repeat area overflow on the source address occurs while SARIE bit is set to 1, the DMCNT.DTE bit is cleared to 0. At the same time, the DMSTS.ESIF flag is set to 1 to indicate that an interrupt by an extended repeat area overflow on the source address is requested.

When block transfer mode is used with the extended repeat area function, an interrupt is requested after completion of a 1-block size transfer. When setting 1 in the DMCNT.DTE bit of the channel for which a transfer has been stopped, the transfer is resumed from the state when the transfer is stopped.

When the extended repeat area is not specified for the source address, this bit is ignored.

Do not set this bit to 1 in repeat-block transfer mode.

RPTIE bit (Repeat Size End Interrupt Enable)

When RPTIE bit is set to 1 in repeat transfer mode, the DMCNT.DTE bit is cleared to 0 after completion of a 1-repeat size data transfer. At the same time, the DMSTS.ESIF flag is set to 1 to indicate that the repeat size end interrupt request has been generated. The repeat size end interrupt request can be generated even when the DMTMD.DTS[1:0] bits are 10b (= repeat area or block area is not specified).

When this bit is set to 1 in block transfer mode, the DMCNT.DTE bit is cleared to 0 after completion of a 1-block data transfer in the same way as repeat transfer mode. At the same time, the DMSTS.ESIF flag is set to 1 to indicate that the repeat size end interrupt request has been generated. The repeat size end interrupt request can be generated even when the DMTMD.DTS[1:0] bits are 10b (= repeat area or block area is not specified).

Do not set this bit to 1 in repeat-block transfer mode.

ESIE bit (Transfer Escape End Interrupt Enable)

ESIE bit enables or disables the transfer escape end interrupt requests (repeat size end interrupt request and extended repeat area overflow interrupt request) that are generated during DMA transfer.

The transfer escape end interrupt is generated when the DMSTS.ESIF flag is set to 1 with this bit set to 1. The transfer escape end interrupt is cleared by clearing this bit or the DMSTS.ESIF flag to 0.

DTIE bit (Transfer End Interrupt Enable)

DTIE bit enables or disables the transfer end interrupt request to be generated on completion of a specified number of data transfers.

The transfer end interrupt is generated when the DMSTS.DTIF flag is set to 1 with this bit set to 1. The transfer end interrupt is cleared by clearing this bit or the DMSTS.DTIF flag to 0.

17.2.12 DMAMD : DMA Address Mode Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x14

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SM[1:0]		SADR	SARA[4:0]					DM[1:0]		DADR	DARA[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	DARA[4:0]	Destination Address Extended Repeat Area Specifies the extended repeat area on the destination address. For details on the settings, see Table 17.2.	R/W
5	DADR	Destination Address Update Select After Reload 0: Only reloading. 1: Add index after reloading.	R/W
7:6	DM[1:0]	Destination Address Update Mode 0 0: Destination address is fixed. 0 1: Offset addition. 1 0: Destination address is incremented. 1 1: Destination address is decremented.	R/W
12:8	SARA[4:0]	Source Address Extended Repeat Area Specifies the extended repeat area on the source address. For details on the settings, see Table 17.2.	R/W
13	SADR	Source Address Update Select After Reload 0: Only reloading. 1: Add index after reloading.	R/W

Bit	Symbol	Function	R/W
15:14	SM[1:0]	Source Address Update Mode 0 0: Source address is fixed. 0 1: Offset addition. 1 0: Source address is incremented. 1 1: Source address is decremented.	R/W

Note: S-TYPE-3, P-TYPE-3

Set DMAMD while DMAC activation is disabled (DMAST.DMST = 0) or DMA transfer is disabled (DMCNT.DTE = 0).

DARA[4:0] bits (Destination Address Extended Repeat Area)

DARA[4:0] bits specify the extended repeat area on the destination address. The extended repeat area function is realized by updating the specified lower address bits with the remaining upper address bits fixed. The size of the extended repeat area can be any power of two between 2 bytes and 128 MB.

When the lower address overflows the extended repeat area by address increment, the start address of the extended repeat area is set. Similarly, when the lower address underflows the extended repeat area by address decrement, the end address of the extended repeat area is set.

When the repeat area or block area is specified as a transfer destination, do not specify the extended repeat area on the destination address. When repeat transfer or block transfer is selected, and when DMTMD.DTS[1:0] = 00b (the transfer destination is specified as the repeat area or block area), write 00000b in the DARA[4:0] bits.

In repeat-block transfer mode, write 00000b in the DARA[4:0] bits.

An interrupt can be requested when an overflow or underflow occurs in the extended repeat area with the DMINT.DARIE bit set to 1. Table 17.2 lists the settings and the corresponding extended repeat areas.

DADR bits (Destination Address Update Select After Reload)

In repeat-block transfer mode, this bit specifies the behavior of DMDAR after reloading DMDRR.

When this bit is set to 1, an index value ((DMDBSH-DMDBSL) × DataSize) is added to DMDAR after reloading DMDRR.

When this bit is set to 0, DMDAR only reloads DMDRR. This behavior is described in Table 17.13.

In normal, repeat or block transfer mode, this bit is ignored.

DM[1:0] bits (Destination Address Update Mode)

DM[1:0] bits select the mode of updating the destination address.

When increment is selected and the DMTMD.SZ[1:0] bits are set to 00b, 01b, 10b, or 11b, the destination address is incremented by 1, 2, 4, or 8 respectively.

When decrement is selected and the DMTMD.SZ[1:0] bits are set to 00b, 01b, 10b, or 11b, the destination address is decremented by 1, 2, 4, or 8 respectively.

When offset addition is selected, the offset specified by the DMOFR register is added to the address.

SARA[4:0] bits (Source Address Extended Repeat Area)

SARA[4:0] bits specify the extended repeat area on the source address. The extended repeat area function is realized by updating the specified lower address bits with the remaining upper address bits fixed. The size of the extended repeat area can be any power of two between 2 bytes and 128 MB.

When the lower address overflows the extended repeat area by address increment, the start address of the extended repeat area is set. Similarly, when the lower address underflows the extended repeat area by address decrement, the end address of the extended repeat area is set.

When the repeat area or block area is specified as a transfer source, do not specify the extended repeat area on the source address. When repeat transfer or block transfer is selected, and when DMTMD.DTS[1:0] = 01b (the transfer source is specified as the repeat area or block area), write 00000b in the SARA[4:0] bits.

In repeat-block transfer mode, write 00000b in the SARA[4:0] bits.

An interrupt can be requested when an overflow or underflow occurs in the extended repeat area with the DMINT.SARIE bit set to 1. Table 17.2 lists the settings and the corresponding extended repeat areas.

SADR bits (Source Address Update Select After Reload)

In repeat-block transfer mode, this bit specifies the behavior of DMSAR after reloading DMSRR.

When this bit is set to 1, an index value ((DMSBSH-DMSBSL) × DataSize) is added to DMSAR after reloading DMSRR.

When this bit is set to 0, DMSAR only reloads DMSRR. This behavior is described in Table 17.12.

In normal, repeat or block transfer mode, this bit is ignored.

SM[1:0] bits (Source Address Update Mode)

SM[1:0] bits select the mode of updating the source address.

When increment is selected and the DMTMD.SZ[1:0] bits are set to 00b, 01b, 10b, or 11b, the source address is incremented by 1, 2, 4, or 8 respectively.

When decrement is selected and the DMTMD.SZ[1:0] bits are set to 00b, 01b, 10b, or 11b, the source address is decremented by 1, 2, 4, or 8 respectively.

When offset addition is selected, the offset specified by the DMOFR register is added to the address.

Table 17.2 SARA[4:0] or DARA[4:0] settings and corresponding repeat areas

SARA[4:0] or DARA[4:0] settings	Extended repeat area
00000b	Not specified
00001b	2 bytes specified as extended repeat area by the lower 1 bit of the address
00010b	4 bytes specified as extended repeat area by the lower 2 bits of the address
00011b	8 bytes specified as extended repeat area by the lower 3 bits of the address
00100b	16 bytes specified as extended repeat area by the lower 4 bits of the address
00101b	32 bytes specified as extended repeat area by the lower 5 bits of the address
00110b	64 bytes specified as extended repeat area by the lower 6 bits of the address
00111b	128 bytes specified as extended repeat area by the lower 7 bits of the address
01000b	256 bytes specified as extended repeat area by the lower 8 bits of the address
01001b	512 bytes specified as extended repeat area by the lower 9 bits of the address
01010b	1 KB specified as extended repeat area by the lower 10 bits of the address
01011b	2 KB specified as extended repeat area by the lower 11 bits of the address
01100b	4 KB specified as extended repeat area by the lower 12 bits of the address
01101b	8 KB specified as extended repeat area by the lower 13 bits of the address
01110b	16 KB specified as extended repeat area by the lower 14 bits of the address
01111b	32 KB specified as extended repeat area by the lower 15 bits of the address
10000b	64 KB specified as extended repeat area by the lower 16 bits of the address
10001b	128 KB specified as extended repeat area by the lower 17 bits of the address
10010b	256 KB specified as extended repeat area by the lower 18 bits of the address
10011b	512 KB specified as extended repeat area by the lower 19 bits of the address
10100b	1 MB specified as extended repeat area by the lower 20 bits of the address
10101b	2 MB specified as extended repeat area by the lower 21 bits of the address
10110b	4 MB specified as extended repeat area by the lower 22 bits of the address
10111b	8 MB specified as extended repeat area by the lower 23 bits of the address
11000b	16 MB specified as extended repeat area by the lower 24 bits of the address
11001b	32 MB specified as extended repeat area by the lower 25 bits of the address
11010b	64 MB specified as extended repeat area by the lower 26 bits of the address
11011b	128 MB specified as extended repeat area by the lower 27 bits of the address
11100b to 11111b	Setting prohibited.

17.2.13 DMOFR : DMA Offset Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x18

Bit position:	31 to 0
Bit field:	
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	n/a	Specifies the offset when offset addition is selected as the address update mode for transfer source or destination. 0x00000000 to 0x00FFFFFF (0 bytes to (16 M – 1) bytes) 0xFF000000 to 0xFFFFFFFF (–16 MB to –1 byte)	R/W

Note: S-TYPE-3, P-TYPE-3

Set DMOFR while DMAC activation is disabled (DMAST.DMST = 0) or DMA transfer is disabled (DMCNT.DTE = 0).

Write to this register while the DMAC operation is stopped or DMA transfer is disabled (not during data transfer).

Setting bits 31 to 25 is invalid; a value of bit 24 is extended to bits 31 to 25. Reading DMOFR returns the extended value.

In repeat-block transfer mode, the offset is not specified by DMOFR when offset addition is selected, write 0 to DMOFR.

17.2.14 DMCNT : DMA Transfer Enable Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x1C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	DTE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DTE	DMA Transfer Enable 0: Disables DMA transfer. 1: Enables DMA transfer.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

DTE bit (DMA Transfer Enable)

When the DMAST.DMST bit is set to 1 (DMAC activation is enabled) and this bit is set to 1 (DMA transfer is enabled), DMA transfer can be started for the corresponding channel.

[Setting condition]

- When 1 is written to this bit.

[Clearing conditions]

- When 0 is written to this bit.
- When the specified total volume of data transfer is completed.
- When DMA transfer is stopped by the repeat size end interrupt.
- When DMA transfer is stopped by the extended repeat area overflow interrupt.
- When DMA transfer is stopped by the access error occurs. See section 17.5. Processing on DMA Transfer Error.

If the DTE of the corresponding channel is set to 0 during DMA transfer, no new transfer requests will be accepted after the transfer ends.

17.2.15 DMREQ : DMA Software Start Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x1D

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	CLRS	—	—	—	SWREQ
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SWREQ	DMA Software Start 0: DMA transfer is not requested. 1: DMA transfer is requested.	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
4	CLRS	DMA Software Start Bit Auto Clear Select 0: SWREQ bit is cleared after DMA transfer is started by software. 1: SWREQ bit is not cleared after DMA transfer is started by software.	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

SWREQ bit (DMA Software Start)

When 1 is written to SWREQ bit, a DMA transfer request is generated. After DMA transfer is started in response to the request, this bit is cleared to 0 if the CLRS bit is set to 0. This bit is not cleared to 0 while the CLRS bit is set to 1. In this case, a DMA transfer request can be issued again after completion of a transfer.

Note that, however, setting this bit is valid and DMA transfer by software is enabled only when the DMTMD.DCTG[1:0] bits are set to 00b (DMAC activation source is software).

Setting this bit is invalid when the DMTMD.DCTG[1:0] bits are set to a value other than 00b.

To start DMA transfer by software with the CLRS bit being 0, ensure that the SWREQ bit is 0, and then write 1 to the SWREQ bit.

[Setting condition]

- When 1 is written to this bit.

[Clearing conditions]

- When a DMA transfer request by software is accepted and DMA transfer is started while the CLRS bit is set to 0 (the SWREQ bit is cleared after DMA transfer is started by software).
- When 0 is written to this bit.

When DMA transfer is stopped by the access error occurs. See section 17.5. Processing on DMA Transfer Error

CLRS bit (DMA Software Start Bit Auto Clear Select)

CLRS bit specifies whether to clear the SWREQ bit to 0 after DMA transfer is started in response to the DMA transfer request generated by setting the SWREQ bit to 1. With this bit set to 0, the SWREQ bit is cleared to 0 after DMA transfer is started. With this bit set to 1, the SWREQ bit is not cleared to 0. In this case, a DMA transfer request can be issued again after completion of a transfer.

17.2.16 DMSTS : DMA Status Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x1E

Bit position:	7	6	5	4	3	2	1	0
Bit field:	ACT	—	—	DTIF	—	—	—	ESIF
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ESIF	Transfer Escape End Interrupt Flag 0: A transfer escape end interrupt has not been generated. 1: A transfer escape end interrupt has been generated.	R/W*1
3:1	—	These bits are read as 0. The write value should be 0.	R/W
4	DTIF	Transfer End Interrupt Flag 0: A transfer end interrupt has not been generated. 1: A transfer end interrupt has been generated.	R/W*1
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	ACT	DMAC Active Flag 0: DMAC is in the idle state. 1: DMAC is operating.	R

Note: S-TYPE-3, P-TYPE-3
Note 1. Only 0 can be written to clear the flag.

ESIF flag (Transfer Escape End Interrupt Flag)

This flag indicates that the transfer escape end interrupt has been generated.

[Setting conditions]

- When 1-repeat size data transfer is completed in repeat transfer mode with the DMINT.RPTIE bit set to 1.
- When 1-block data transfer is completed in block transfer mode with the DMINT.RPTIE bit set to 1.
- When an extended repeat area overflow on the source address occurs while the DMINT.SARIE bit is set to 1 and the DMAMD.SARA[4:0] bits are set to a value other than 00000b (extended repeat area is specified on the transfer source address).
- When an extended repeat area overflow on the destination address occurs while the DMINT.DARIE bit is set to 1 and the DMAMD.DARA[4:0] bits are set to a value other than 00000b (extended repeat area is specified on the transfer destination address).

[Clearing conditions]

- When 0 is written to this bit.
- When 1 is written to the DMCNT.DTE bit.

DTIF flag (Transfer End Interrupt Flag)

This flag indicates that the transfer end interrupt has been generated.

[Setting conditions]

- When the specified number of unit-transfers are completed in normal transfer mode (the value of DMCRAL becoming 0 on completion of transfer).
- When the specified number of repeat transfer operations are completed in repeat transfer mode (the value of DMCRBL becoming 0 on completion of transfer with DMTMD.TKP = 0 or the value of DMCRBL reloading DMCRBH with DMTMD.TKP = 1).
- When the specified number of blocks have been transferred in block transfer mode and repeat-block transfer mode (the value of DMCRBL becoming 0 on completion of transfer with DMTMD.TKP = 0 or the value of DMCRBL reloading DMCRBH with DMTMD.TKP = 1).

[Clearing conditions]

- When 0 is written to this bit.
- When 1 is written to the DMCNT.DTE bit.

ACT flag (DMAC Active Flag)

This flag indicates whether the DMAC is in the idle or active state.

[Setting condition]

- When the DMAC starts data transfer operation.

[Clearing condition]

- When data transfer in response to one transfer request is completed.

17.2.17 DMSBS : DMA Source Buffer Size Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x28

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	DMSBSH[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DMSBSL[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	DMSBSL[15:0]	Functions as data transfer counter in repeat-block transfer mode See Table 17.3 for available settings.	R/W
31:16	DMSBSH[15:0]	Specifies the repeat-area size in repeat-block transfer mode See Table 17.3 for available settings.	R/W

Note: S-TYPE-3, P-TYPE-3

Set DMSBS while DMAC activation is disabled (DMAST.DMST = 0) or DMA transfer is disabled (DMCNT.DTE = 0).

Set the same value for DMSBSH and DMSBSL in repeat-block transfer mode. Write 0x00000000 to DMSBS in normal, repeat and block transfer mode.

DMSBSH specifies buffer size and DMSBSL functions as a 16-bit buffer size counter in repeat-block transfer mode. In repeat-block transfer mode, source repeat area is specified by DMSBSH.

When address update mode is incremented address or decremented address, this register means the numbers of data of whole buffer. When address update mode is offset addition, this register means the numbers of data of an individual buffer. In offset addition, setting DMSBSH and DMSBSL to 0x0000 is prohibited. When final data of one buffer size is transferred, DMSBSL reloads value of DMSBSH. When address update mode is fixed address, this register is ignored. Table 17.3 shows the setting values of DMA Source Buffer Size Register corresponding to Transfer Data Size in Source Address Update Mode.

Table 17.3 Available setting for DMSBS register in repeat-block transfer mode (1 of 2)

Source address update mode (DMAMD.SM)	Transfer data size (DMTMD.SZ)	Available setting for DMSBSH and DMSBSL bits
Source address is fixed (SM = 00b)	Don't care	0x0000 (DMSBS is not used)

Table 17.3 Available setting for DMSBS register in repeat-block transfer mode (2 of 2)

Source address update mode (DMAMD.SM)	Transfer data size (DMTMD.SZ)	Available setting for DMSBSH and DMSBSL bits
Offset addition (SM = 01b)	8 bits (SZ = 00b)	0x0001 to 0xFFFF (1 to 65535)
	16 bits (SZ = 01b)	0x0001 to 0x7FFF (1 to 32767)
	32 bits (SZ = 10b)	0x0001 to 0x3FFF (1 to 16383)
	64 bits (SZ = 11b)	0x0001 to 0x1FFF (1 to 8191)
Source address is incremented or decremented (SM = 1xb)	Don't care	0x0000 (infinite) 0x0001 to 0xFFFF (1 to 65535)

In normal, repeat and block transfer mode, DMSBS is not used. The setting is invalid.

17.2.18 DMDBS : DMA Destination Buffer Size Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x2C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	DMDBSH[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DMDBSL[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	DMDBSL[15:0]	Functions as data transfer counter in repeat-block transfer mode. See Table 17.4 for available settings.	R/W
31:16	DMDBSH[15:0]	Specifies the repeat-area size in repeat-block transfer mode. See Table 17.4 for available settings.	R/W

Note: S-TYPE-3, P-TYPE-3

Set DMDBS while DMAC activation is disabled (DMAST.DMST = 0) or DMA transfer is disabled (DMCNT.DTE = 0).

Set the same value for DMDBSH and DMDBSL in repeat-block transfer mode. Write 0x00000000 to DMDBS in normal, repeat and block transfer mode.

DMDBSH specifies buffer size and DMDBSL functions as a 16-bit buffer size counter in repeat-block transfer mode. In repeat-block transfer mode, destination repeat area is specified by DMDBSH.

When address update mode is incremented address or decremented address, this register means the numbers of data of whole buffer. When address update mode is offset addition, this register means the numbers of data of an individual buffer. In offset addition, setting DMDBSH and DMDBSL to 0x0000 is prohibited. When final data of one buffer size is transferred, DMDBSL reloads value of DMDBSH. When address update mode is fixed address, this register is ignored. Table 17.4 shows the setting values of Destination Buffer Size Register corresponding to Transfer Data Size in Destination Address Update Mode.

Table 17.4 Available setting for DMDBS register in repeat-block transfer mode (1 of 2)

Destination address update mode (DMAMD.DM)	Transfer data size (DMTMD.SZ)	Available setting for DMDBSH and DMDBSL bits
Destination address is fixed (DM = 00b)	Don't care	0x0000 (DMDBS is not used)

Table 17.4 Available setting for DMDBS register in repeat-block transfer mode (2 of 2)

Destination address update mode (DMAMD.DM)	Transfer data size (DMTMD.SZ)	Available setting for DMDBSH and DMDBSL bits
Offset addition (DM = 01b)	8 bits (SZ = 00b)	0x0001 to 0xFFFF (1 to 65535)
	16 bits (SZ = 01b)	0x0001 to 0x7FFF (1 to 32767)
	32 bits (SZ = 10b)	0x0001 to 0x3FFF (1 to 16383)
	64 bits (SZ = 11b)	0x0001 to 0x1FFF (1 to 8191)
Destination address is incremented or decremented (DM = 1xb)	Don't care	0x0000 (infinite) 0x0001 to 0xFFFF (1 to 65535)

In normal, repeat and block transfer mode, DMDBS is not used. The setting is invalid.

17.2.19 DMBWR : DMA Bufferable Write Enable Register

Base address: DMAC0n = 0x4000_A000 + 0x0040 × n (n = 0 to 7)
DMAC0n_NS = 0x5000_A000 + 0x0040 × n (n = 0 to 7)

Offset address: 0x30

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	BWE
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BWE	Bufferable Write Enable 0: Disables Bufferable Write 1: Enables Bufferable Write	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Set DMBWR while DMAC activation is disabled (DMAST.DMST = 0) or DMA transfer is disabled (DMCNT.DTE = 0).

BWE bit (Bufferable Write Enable)

BWE bit indicates bufferable write is either enabled or disabled.

If this bit is 1, even if the write access as DMAC is completed, the actual slave write may not have ended.

When this bit is 1, it requests an early response by writing write data to the temporary buffer if possible to the slave. Therefore, even if write access as DMAC is completed, the actual slave write may not be finished. If an error occurs during write access to the target slave, the error may not be detected and the transfer may not be automatically stopped. In such cases, the error response detection interrupt DMAm_TRANSERR (m = 0, 1) will not occur.

See section 15.7.2. Operations When a Bus Error Occurs for slave groups that support bufferable writes and slave groups for which error response detection interrupts cannot be generated.

Some individual slave modules also support bufferable writes, so Refer to the respective module chapters.

[Setting condition]

- When 1 is written to this bit

[Clearing conditions]

- When 0 is written to this bit

17.2.20 DMAST : DMA Module Activation Register

Base address: DMA0 = 0x4000_A800
DMA0_NS = 0x5000_A800

Offset address: 0x00

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	DMST
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DMST	DMAC Operation Enable 0: DMAC activation is disabled. 1: DMAC activation is enabled.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE2

DMST bit (DMAC Operation Enable)

Setting the DMAST.DMST to 1 enables DMAC activation for all channels. When the DMST bit is set to 1 (DMAC activation is enabled), and 1 is written to the DMCNT.DTE bit (DMA transfer is enabled) for multiple channels, all associated channels can be placed in the transfer request ready state at the same time.

When the DMST bit clears to 0 during DMA transfer, DMA transfer is suspended after the current data transfer associated with a single transfer request completes. To resume DMA transfer, set the DMST bit to 1 again.

[Setting condition]

- When 1 is written to this bit.

[Clearing condition]

- When 0 is written to this bit.

17.2.21 DMECHR : DMAC Error Channel Register

Base address: DMA0 = 0x4000_A800
DMA0_NS = 0x5000_A800

Offset address: 0x40

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DMESTA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	DMECHSAM	—	—	—	—	DMECH[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	DMECH[3:0]	DMAC Error Channel Indicates the channel number causing the error 0000: Error occurred on channel 0 0001: Error occurred on channel 1 0010: Error occurred on channel 2 ⋮ ⋮ 0111: Error occurred on channel 7	R
7:4	—	These bits are read as 0. The write value should be 0.	R

Bit	Symbol	Function	R/W
8	DMECHSAM	DMAC Error Channel Security Attribution Monitor Indicates the security attribution of a channel causing the error 0: secure channel 1: non-secure channel	R
15:9	—	These bits are read as 0. The write value should be 0.	R
16	DMESTA	DMAC Error Status 0: No DMA transfer error occurred 1: DMA transfer error occurred	R/W *1
31:17	—	These bits are read as 0. The write value should be 0.	R

Note: P-TYPE-2

Note 1. Writing to DMESTA depends on the value of DMECHSAM.

When reading this register, it can be accessed from both secure and non-secure access.

When writing this register, it depends on DMECHR.DMESA.

- When DMESA = 1, it can be accessed from secure and non-secure access.
- When DMESA = 0, it can be accessed from secure. An error is returned when write access is performed in a non-secure access.

This register is cleared by a reset caused by a transfer error. Please select interrupt DMAm_TRANSERR (m = 0, 1) in BUS.OADCFG.OAD when you want to debug the program.

DMECH[3:0] bits (DMAC Error Channel)

When a transfer error due to DMA transfer occurs, it stores the channel of DMAC that was violated.

When reset was selected in MPU.MMPUOAD.OAD and TZF.TZFOAD.OAD, Since this register is also reset. Please select NMI when you want to debug the program.

[Set condition]

- When the DMAC transfer error occurs and DMESTA = 0.

[Clearing condition]

- When 1 is written to DMESTA.

DMECHSAM bit (DMAC Error Channel Security Attribution Monitor)

When a transfer error due to DMA transfer occurs, it indicates the security attribution of the violating DMAC channel.

When reset was selected in MPU.MMPUOAD.OAD and TZF.TZFOAD.OAD, Since this register is also reset. Please select NMI when you want to debug the program.

[Set condition]

- When the DMAC transfer error occurs and DMESTA = 0.

[Clearing condition]

- When 1 is written to DMESTA.

DMESTA bit (DMAC Error Status)

Indicates whether or not a DMA transfer error occurred.

DMECH[3:0], DMECHSAM, DMESTA are cleared by writing 1 to DMESTA. Writing 0 to DMESTA is ignored.

When reset was selected in MPU.MMPUOAD.OAD and TZF.TZFOAD.OAD, Since this register is also reset. Please select NMI when you want to debug the program.

[Set condition]

- When the DMAC transfer error occurs.

[Clearing condition]

- When 1 is written to DMESTA.

Note: When DMECHSAM = 1, it can be cleared in the secure state and non-secure state. DMECHSAM = 0, it cannot be cleared in the non-secure state.

17.2.22 DMCTL : DMAC Control Register

Base address: DMA0 = 0x4000_A800
DMA0_NS = 0x5000_A800

Offset address: 0x10

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	ERCH	—	—	—	PR
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PR	Priority Control Select 0: Fixed priority mode 1: Round-robin mode	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
4	ERCH	Clear Channel Select 0: Target channel only 1: All channels	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-2

Set DMCTL while DMAC activation is disabled (DMAST.DMST = 0) or DMA transfer is disabled (DMCNT.DTE = 0).

PR bit (Priority Control Select)

When DMCTL.PR = 0, arbitration is performed in the fixed priority mode of channel 0 > channel 1 > channel 2 > channel 3 …> channel 7 > DTC (channel 0: highest priority) and DMA transfer is started.

When DMCTL.PR = 1, arbitration is performed in the round-robin mode and DMA transfer is started.

ERCH bit (Clear Channel Select)

When DMCTL.ERCH = 0, the ICU clears all bits of the target channel of DELSRn (n = 0 to 7) by the error response during transfer. DELSRn (n = 0 to 7) other than the target channel is not cleared.

When DMCTL.ERCH = 1, the ICU clears all bits of all channels of DELSRn (n = 0 to 7) by the error response during transfer.

17.3 Operation

17.3.1 Transfer Mode

17.3.1.1 Normal Transfer Mode

In normal transfer mode, one data is transferred by one transfer request. A maximum of 65535 can be set as the number of transfer operations using the DMCRAL register. When these bits are set to 0x0000, no specific number of transfer operations is set; data transfer is performed with the transfer counter stopped (free-running function). Setting DMCRB register is invalid in normal transfer mode. Except in free-running function, a transfer end interrupt request can be generated after completion of the specified number of transfer operations.

Table 17.5 summarizes the register update operation in normal transfer mode, and Figure 17.2 shows the operation in normal transfer mode.

Table 17.5 Register update operation in normal transfer mode

Register	Function	Update operation after completion of a transfer by one transfer request
DMSAR	Transfer source address	Increment/decrement/fixed/offset addition
DMDAR	Transfer destination address	Increment/decrement/fixed/offset addition
DMCRAL	Transfer count	Decremented by one/not updated (in free running function)
DMCRAH	—	Not updated (Not used in normal transfer mode)
DMCRB	—	Not updated (Not used in normal transfer mode)

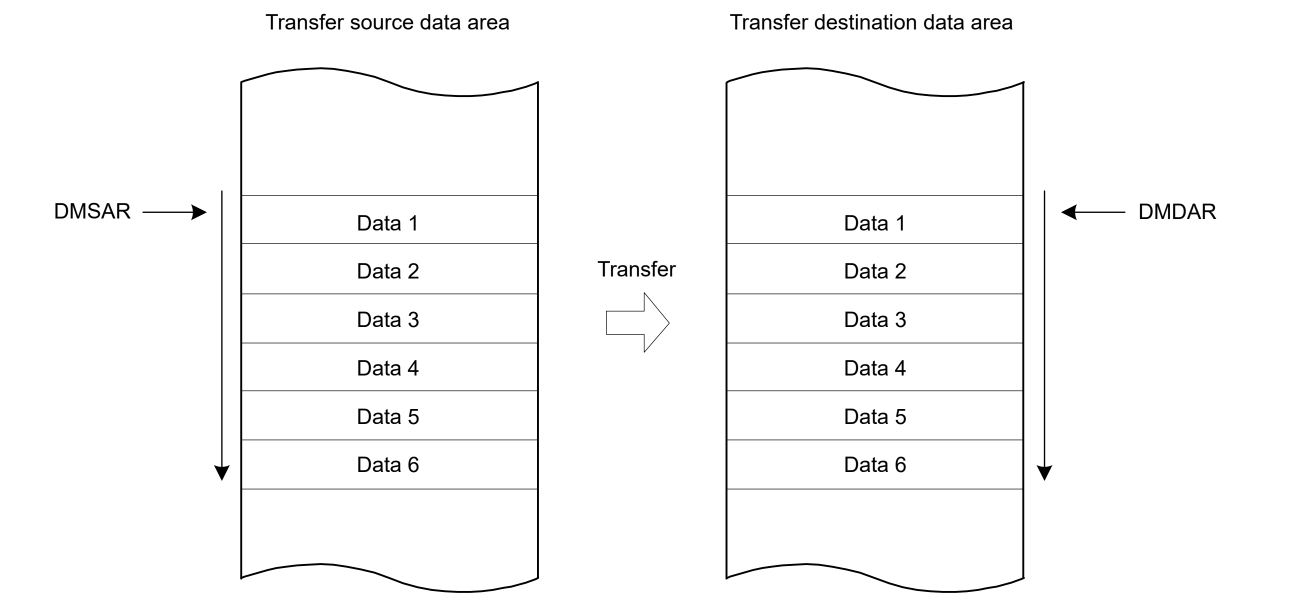

Figure 17.2 Operation in normal transfer mode

17.3.1.2 Repeat Transfer Mode

In repeat transfer mode, one data is transferred by one transfer request.

A maximum of 1K data can be set as a total repeat transfer size using DMCRA register.

A maximum of 64K can be set as the number of repeat transfer operations using DMCRB register; therefore, a maximum of 64M data (1K data × 64K counts of repeat transfer operations) can be set as a total data transfer size.

Either the transfer source or transfer destination can be specified as a repeat area. When transfer of the repeat size data is completed, the address of the specified repeat area (DMSAR or DMDAR) returns to the transfer start address. When data of the specified repeat size has all been transferred in repeat transfer mode, DMA transfer can be stopped, and the repeat size end interrupt can be requested. DMA transfer can be resumed by writing 1 to the DMCNT.DTE bit in the repeat size end interrupt handling.

A transfer end interrupt request can be generated after completion of the specified number of repeat transfer operations.

Table 17.6 summarizes the register update operation in repeat transfer mode, and Figure 17.3 shows the operation in repeat transfer mode.

Table 17.6 Register update operation in repeat transfer mode

Register	Function	Update operation after completion of a transfer by one transfer request: When DMCRAL register is not 1	Update operation after completion of a transfer by one transfer request: When DMCRAL register is 1 (Transfer of the last data in repeat size)
DMSAR	Transfer source address	Increment/decrement/fixed/offset addition	• DMTMD.DTS[1:0] = 00b Increment/decrement/fixed/offset addition • DMTMD.DTS[1:0] = 01b Initial value of DMSAR • DMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition
DMDAR	Transfer destination address	Increment/decrement/fixed/offset addition	• DMTMD.DTS[1:0] = 00b Initial value of DMDAR • DMTMD.DTS[1:0] = 01b Increment/decrement/fixed/offset addition • DMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition
DMCRAH	Repeat size	Not updated	Not updated
DMCRAL	Transfer count	Decremented by one	DMCRAH
DMCRBH	Number of repeat transfer operations	Not updated	Not updated
DMCRBL	Count of repeat transfer operations	Not updated	Decremented by one

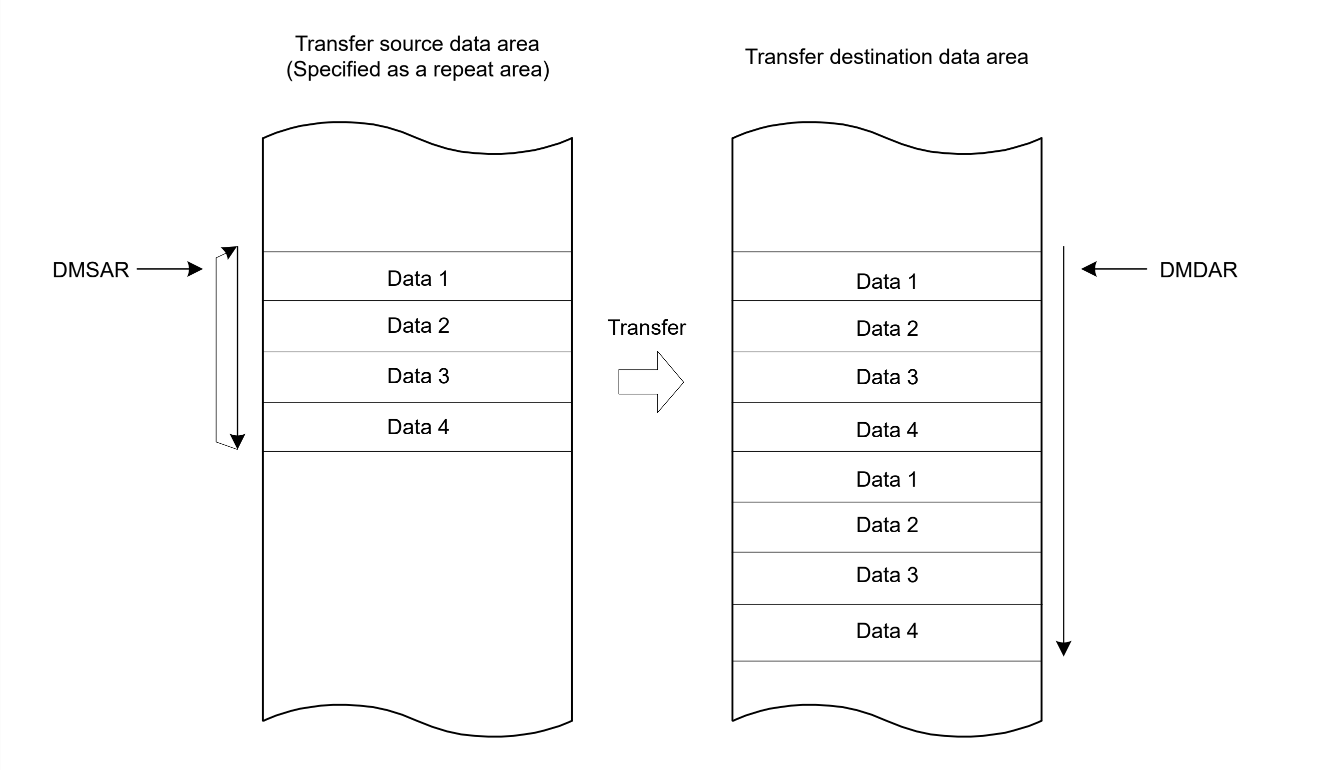

Figure 17.3 Operation in repeat transfer mode

17.3.1.3 Block Transfer Mode

In block transfer mode, a single block data is transferred by one transfer request.

A maximum of 1K data can be set as a total block transfer size using DMCRA register.

A maximum of 64K can be set as the number of block transfer operations using DMCRB register; therefore, a maximum of 64M data (1K data × 64K counts of block transfer operations) can be set as a total data transfer size.

Either the transfer source or transfer destination can be specified as a block area. When transfer of a single block data is completed, the address of the specified block area (DMSAR or DMDAR) returns to the transfer start address. When a single block data has all been transferred in block transfer mode, DMA transfer can be stopped, and the repeat size end interrupt can be requested. DMA transfer can be resumed by writing 1 to the DMCNT.DTE bit in the repeat size end interrupt handling.

Transfer end interrupt request can be generated after completion of the specified number of block transfer operations.

Table 17.7 summarizes the register update operation in block transfer mode, and Figure 17.4 shows the operation in block transfer mode.

Table 17.7 Register update operation in block transfer mode

Register	Function	Update operation after completion of single-block transfer by one transfer request
DMSAR	Transfer source address	• DMTMD.DTS[1:0] = 00b Increment/decrement/fixed/offset addition • DMTMD.DTS[1:0] = 01b Initial value of DMSAR • DMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition
DMDAR	Transfer destination address	• DMTMD.DTS[1:0] = 00b Initial value of DMDAR • DMTMD.DTS[1:0] = 01b Increment/decrement/fixed/offset addition • DMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition
DMCRAH	Block size	Not updated
DMCRAL	Transfer count	DMCRAH
DMCRBH	Number of block transfer operations	Not updated
DMCRBL	Count of block transfer operations	Decremented by one

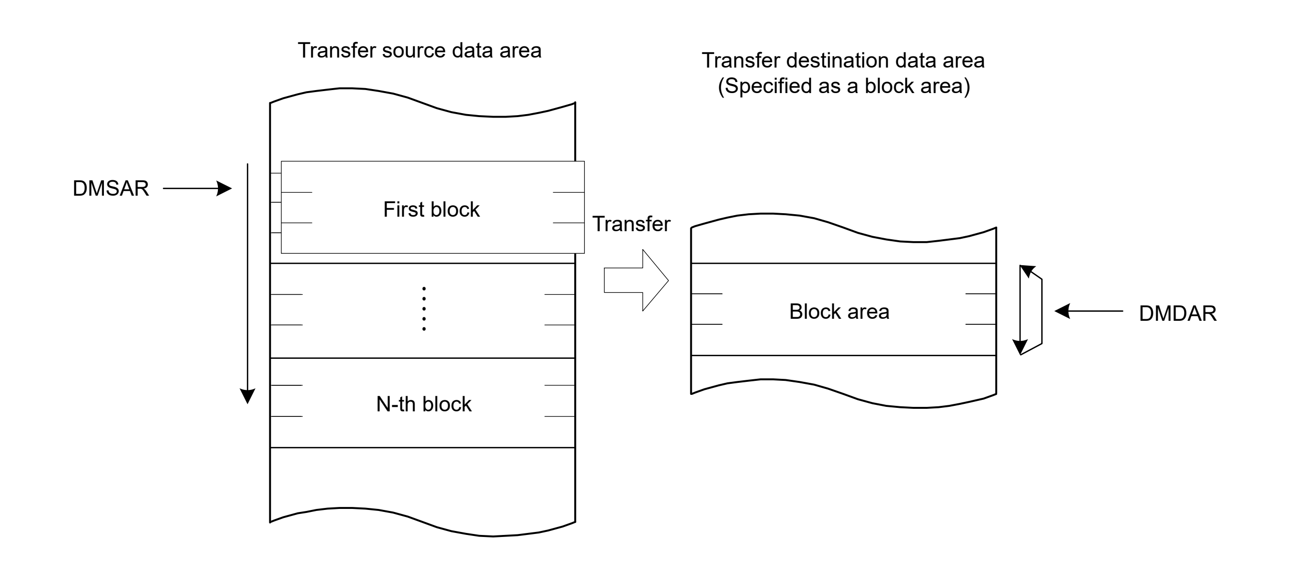

Figure 17.4 Operation in block transfer mode

17.3.1.4 Repeat-Block Transfer Mode

Repeat-block transfer is the operation mode with the following functions added to the block transfer function.

Repeat function: Added function (ring buffer) to repeat specified address area.

Offset function: Multiple areas with offset can be specified within one block transfer.

The repeat function and the offset function can be used for both the transfer source and the transfer destination of repeat-block transfer.

Figure 17.5 shows an example of adding a repeat function to the transfer destination.

Figure 17.6 shows repeat-block transfer with an offset to the transfer destination.

In repeat-block transfer mode, a single block data is transferred by one transfer request.

A maximum of 1K data can be set as a total block transfer size using DMCRA of the DMACn.

A maximum of 64K can be set as the number of block transfer operations using DMCRB of the DMACn; therefore, a maximum of 64M data (1K data × 64K counts of block transfer operations) can be set as a total data transfer size.

A transfer end interrupt request can be generated after completion of the specified number of repeat-block transfer operations.

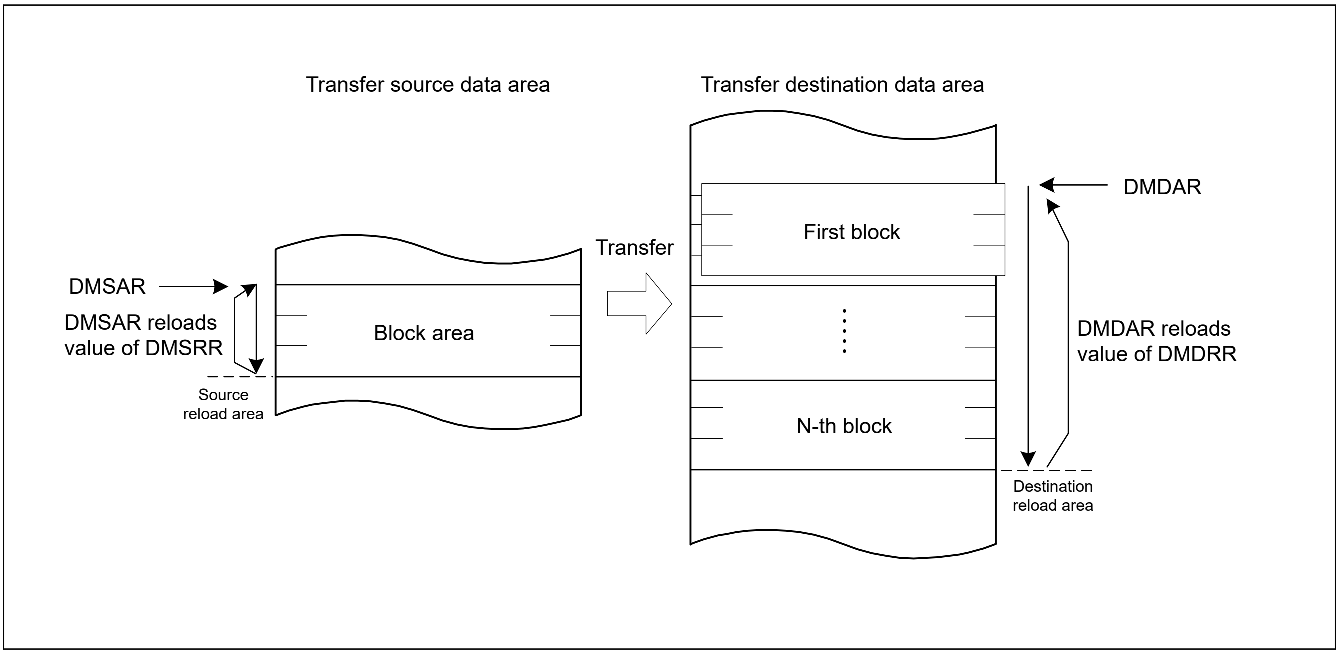

Figure 17.5 Operation in repeat-block transfer mode

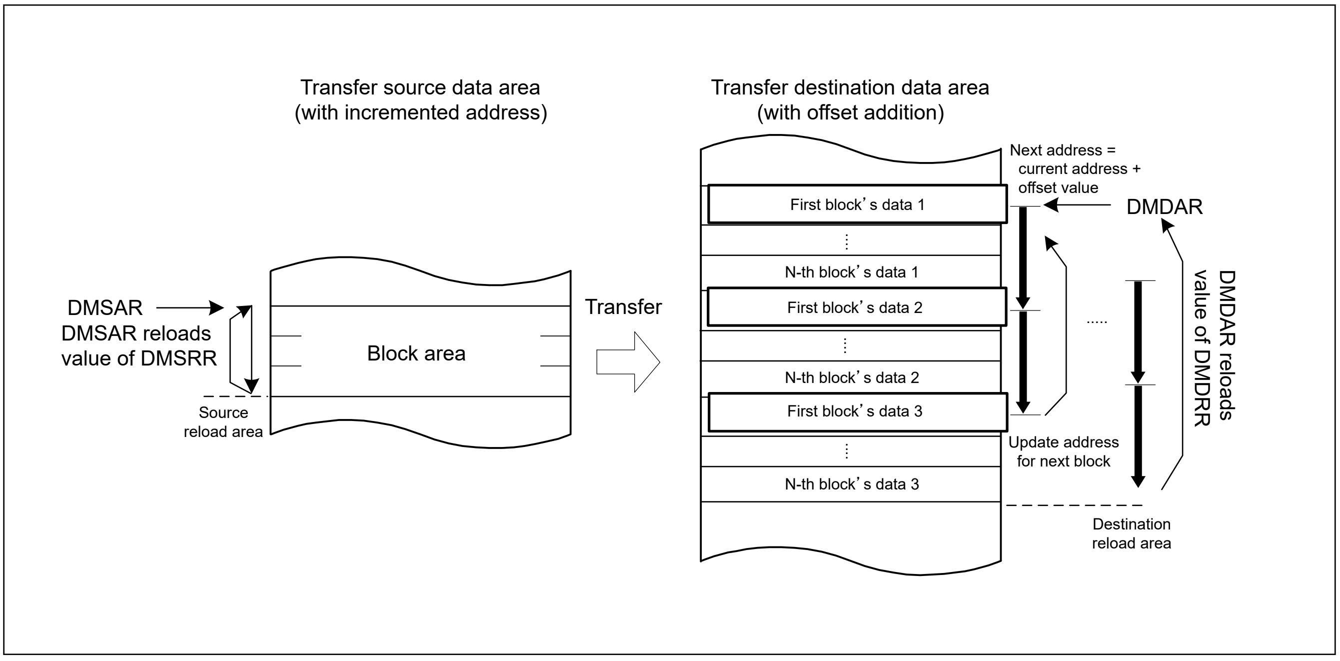

Figure 17.6 Operation in repeat-block transfer mode with offset addition

Table 17.8 to Table 17.13 summarize the register update operations in repeat-block transfer mode.

For more information about address update function in repeat-block transfer mode, see section 17.3.5. Address Update Function in Repeat-Block Transfer Mode.

Table 17.8 Register update operation associated with source area in repeat-block transfer mode (fixed address DMAMD.SM[1:0] = 00b)

<table>
<tr><th rowspan="3">Register</th><th rowspan="3">Function</th><th colspan="3">Update operation after single data is transferred</th></tr>
<tr><th rowspan="2">DMCRAL[15:0] is not 1</th><th colspan="2">DMCRAL[15:0] is 1 (single block is transferred)</th></tr>
<tr><th>DMCRBL[15:0] is not 1</th><th>DMCRBL[15:0] is 1</th></tr>
<tr><td>DMSRR</td><td>Transfer source reload address</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td>DMSAR</td><td>Transfer source address</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td>DMCRAH[9:0]</td><td>Block size</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td>DMCRAL[15:0]</td><td>Block size count</td><td>Decremented by 1</td><td>DMCRAH[9:0]</td><td>DMCRAH[9:0]</td></tr>
<tr><td>DMCRBH[15:0]</td><td>Number of block transfer operations</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td rowspan="2">DMCRBL[15:0]</td><td>Count of block transfer operations when DMTMD.TKP = 0</td><td rowspan="2">Not updated</td><td rowspan="2">Decremented by 1</td><td>0</td></tr>
<tr><td>Count of block transfer operations when DMTMD.TKP = 1</td><td>DMCRBH[15:0]</td></tr>
</table>

Table 17.9 Register update operation associated with destination area in repeat-block transfer mode (fixed address DMAMD.DM[1:0] = 00b)

<table>
<tr><th rowspan="3">Register</th><th rowspan="3">Function</th><th colspan="3">Update operation after single data is transferred</th></tr>
<tr><th rowspan="2">DMCRAL[15:0] is not 1</th><th colspan="2">DMCRAL[15:0] is 1 (single block is transferred)</th></tr>
<tr><th>DMCRBL[15:0] is not 1</th><th>DMCRBL[15:0] is 1</th></tr>
<tr><td>DMDRR</td><td>Transfer destination reload address</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td>DMDAR</td><td>Transfer destination address</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td>DMCRAH[9:0]</td><td>Block size</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td>DMCRAL[15:0]</td><td>Block size count</td><td>Decremented by 1</td><td>DMCRAH[9:0]</td><td>DMCRAH[9:0]</td></tr>
<tr><td>DMCRBH[15:0]</td><td>Number of block transfer operations</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td rowspan="2">DMCRBL[15:0]</td><td>Count of block transfer operations when DMTMD.TKP = 0</td><td rowspan="2">Not updated</td><td rowspan="2">Decremented by 1</td><td>0</td></tr>
<tr><td>Count of block transfer operations when DMTMD.TKP = 1</td><td>DMCRBH[15:0]</td></tr>
</table>

Table 17.10 Register update operation associated with source area in repeat-block transfer mode (incremented or decremented address DMAMD.SM[1:0] = 10b or 11b)

<table>
<tr><th rowspan="4">Register</th><th rowspan="4">Function</th><th colspan="6">Update operation after single data is transferred</th></tr>
<tr><th colspan="3">DMSBSL[15:0] is not 1</th><th colspan="3">DMSBSL[15:0] is 1</th></tr>
<tr><th rowspan="2">DMCRAL [15:0] is not 1</th><th colspan="2">DMCRAL[15:0] is 1 (single block is transferred)</th><th rowspan="2">DMCRAL [15:0] is not 1</th><th colspan="2">DMCRAL[15:0] is 1 (single block is transferred)</th></tr>
<tr><th>DMCRBL[15:0] is not 1</th><th>DMCRBL[15:0] is 1</th><th>DMCRBL[15:0] is not 1</th><th>DMCRBL[15:0] is 1</th></tr>
<tr><td>DMSRR</td><td>Transfer source reload address</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td rowspan="2">DMSAR</td><td>Transfer source address when DMTMD.SM [1:0] = 10b</td><td colspan="3">Incremented by Data Size</td><td colspan="3">DMSRR</td></tr>
<tr><td>Transfer source address when DMTMD.SM [1:0] = 11b</td><td colspan="3">Decremented by Data Size</td><td colspan="3">DMSRR</td></tr>
<tr><td>DMCRAH[9:0]</td><td>Block size</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td>DMCRAL[15:0]</td><td>Block size count</td><td>Decremented by 1</td><td>DMCRAH[9:0]</td><td>DMCRAH[9:0]</td><td>Decremented by 1</td><td>DMCRAH[9:0]</td><td>DMCRAH[9:0]</td></tr>
<tr><td>DMSBSH[15:0]</td><td>Source buffer size (Repeat-size)</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td>DMSBSL[15:0]</td><td>Count of transfer data in source buffer</td><td>Decremented by 1</td><td>Decremented by 1</td><td>Decremented by 1</td><td>DMSBSH</td><td>DMSBSH</td><td>DMSBSH</td></tr>
<tr><td>DMCRBH[15:0]</td><td>Number of block transfer operations</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td rowspan="2">DMCRBL[15:0]</td><td>Count of block transfer operations when DMTMD.TKP = 0</td><td rowspan="2">Not updated</td><td rowspan="2">Decremented by 1</td><td>0</td><td rowspan="2">Not updated</td><td rowspan="2">Decremented by 1</td><td>0</td></tr>
<tr><td>Count of block transfer operations when DMTMD.TKP = 1</td><td>DMCRBH[15:0]</td><td>DMCRBH[15:0]</td></tr>
</table>

Table 17.11 Register update operation associated with destination area in repeat-block transfer mode (incremented or decremented address DMAMD.DM[1:0] = 10b or 11b)

<table>
<tr><th rowspan="4">Register</th><th rowspan="4">Function</th><th colspan="6">Update operation after single data is transferred</th></tr>
<tr><th colspan="3">DMDBSL[15:0] is not 1</th><th colspan="3">DMDBSL[15:0] is 1</th></tr>
<tr><th rowspan="2">DMCRAL [15:0] is not 1</th><th colspan="2">DMCRAL[15:0] is 1 (single block is transferred)</th><th rowspan="2">DMCRAL [15:0] is not 1</th><th colspan="2">DMCRAL[15:0] is 1 (single block is transferred)</th></tr>
<tr><th>DMCRBL[15:0] is not 1</th><th>DMCRBL[15:0] is 1</th><th>DMCRBL[15:0] is not 1</th><th>DMCRBL[15:0] is 1</th></tr>
<tr><td>DMDRR</td><td>Transfer destination reload address</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td rowspan="2">DMDAR</td><td>Transfer destination address when DMTMD.DM [1:0] = 10b</td><td colspan="3">Incremented by Data Size</td><td colspan="3">DMDRR</td></tr>
<tr><td>Transfer destination address when DMTMD.DM [1:0] = 11b</td><td colspan="3">Decremented by Data Size</td><td colspan="3">DMDRR</td></tr>
<tr><td>DMCRAH[9:0]</td><td>Block size</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td>DMCRAL[15:0]</td><td>Block size count</td><td>Decremented by 1</td><td>DMCRAH[9:0]</td><td>DMCRAH[9:0]</td><td>Decremented by 1</td><td>DMCRAH[9:0]</td><td>DMCRAH[9:0]</td></tr>
<tr><td>DMDBSH[15:0]</td><td>Destination buffer size (Repeat-size)</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td>DMDBSL[15:0]</td><td>Count of transfer data in destination buffer</td><td>Decremented by 1</td><td>Decremented by 1</td><td>Decremented by 1</td><td>DMDBSH</td><td>DMDBSH</td><td>DMDBSH</td></tr>
<tr><td>DMCRBH[15:0]</td><td>Number of block transfer operations</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
<tr><td rowspan="2">DMCRBL[15:0]</td><td>Count of block transfer operations when DMTMD.TKP = 0</td><td rowspan="2">Not updated</td><td rowspan="2">Decremented by 1</td><td>0</td><td rowspan="2">Not updated</td><td rowspan="2">Decremented by 1</td><td>0</td></tr>
<tr><td>Count of block transfer operations when DMTMD.TKP = 1</td><td>DMCRBH[15:0]</td><td>DMCRBH[15:0]</td></tr>
</table>

Table 17.12 Register update operation associated with source area in repeat-block transfer mode (offset addition DMAMD.SM[1:0] = 01b) (1 of 2)

<table>
<tr><th rowspan="3">Register</th><th rowspan="3">Function</th><th rowspan="3">DMCRAL[15:0] is not 1</th><th colspan="4">DMCRAL[15:0] is 1 (single block is transferred)</th></tr>
<tr><th colspan="2">DMSBSL[15:0] is not 1</th><th colspan="2">DMSBSL[15:0] is 1</th></tr>
<tr><th>DMCRBL[15:0] is not 1</th><th>DMCRBL[15:0] is 1</th><th>DMCRBL[15:0] is not 1</th><th>DMCRBL[15:0] is 1</th></tr>
<tr><td>DMSRR</td><td>Transfer source reload address</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td><td>Not updated</td></tr>
</table>

Table 17.12 Register update operation associated with source area in repeat-block transfer mode (offset addition DMAMD.SM[1:0] = 01b) (2 of 2)

Register	Function	DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)			
			DMSBSL[15:0] is not 1		DMSBSL[15:0] is 1	
			DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1	DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1
DMSAR	Transfer source address when DMAMD.SADR = 0	Offset addition by DMSBSH	DMSRR		DMSRR	
	Transfer source address when DMAMD.SADR = 1		DMSRR + (DMS-BSH - DMSBSL) × DataSize			
DMCRAH[9:0]	Block size	Not updated	Not updated	Not updated	Not updated	Not updated
DMCRAL[15:0]	Block size count	Decremented by 1	DMCRAH[9:0]	DMCRAH[9:0]	DMCRAH[9:0]	DMCRAH[9:0]
DMSBSH[15:0]	Source buffer size (Repeat-size)	Not updated	Not updated	Not updated	Not updated	Not updated
DMSBSL[15:0]	Count of transfer data in source buffer	Not updated	Decremented by 1	Decremented by 1	DMSBSH	DMSBSH
DMCRBH[15:0]	Number of block transfer operations	Not updated	Not updated	Not updated	Not updated	Not updated
DMCRBL[15:0]	Count of block transfer operations when DMTMD.TKP = 0	Not updated	Decremented by 1	0	Decremented by 1	0
	Count of block transfer operations when DMTMD.TKP = 1			DMCRBH[15:0]		DMCRBH[15:0]

Table 17.13 Register update operation associated with destination area in repeat-block transfer mode (offset addition DMAMD.DM[1:0] = 01b) (1 of 2)

Register	Function	DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)			
			DMDBSL[15:0] is not 1		DMDBSL[15:0] is 1	
			DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1	DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1
DMDRR	Transfer destination reload address	Not updated	Not updated	Not updated	Not updated	Not updated
DMDAR	Transfer destination address when DMAMD.DADR = 0	Offset addition by DMDBSH	DMDRR		DMDRR	
	Transfer destination address when DMAMD.DADR = 1		DMDRR + (DMDBSH - DMDBSL) × DataSize			
DMCRAH[9:0]	Block size	Not updated	Not updated	Not updated	Not updated	Not updated
DMCRAL[15:0]	Block size count	Decremented by 1	DMCRAH[9:0]	DMCRAH[9:0]	DMCRAH[9:0]	DMCRAH[9:0]

Table 17.13 Register update operation associated with destination area in repeat-block transfer mode (offset addition DMAMD.DM[1:0] = 01b) (2 of 2)

Register	Function	DMCRAL[15:0] is not 1	DMCRAL[15:0] is 1 (single block is transferred)			
			DMDBSL[15:0] is not 1		DMDBSL[15:0] is 1	
			DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1	DMCRBL[15:0] is not 1	DMCRBL[15:0] is 1
DMDBSH[15:0]	Destination buffer size (Repeat-size)	Not updated	Not updated	Not updated	Not updated	Not updated
DMDBSL[15:0]	Count of transfer data in destination buffer	Not updated	Decremented by 1	Decremented by 1	DMDBSH	DMDBSH
DMCRBH[15:0]	Number of block transfer operations	Not updated	Not updated	Not updated	Not updated	Not updated
DMCRBL[15:0]	Count of block transfer operations when DMTMD.TKP = 0	Not updated	Decremented by 1	0	Decremented by 1	0
	Count of block transfer operations when DMTMD.TKP = 1			DMCRBH[15:0]		DMCRBH[15:0]

17.3.2 Extended Repeat Area Function

The DMAC supports a function to specify the extended repeat areas on the transfer source and destination addresses. With the extended repeat areas set, the address registers repeatedly indicate the addresses of the specified extended repeat areas.

The extended repeat areas can be specified separately to the transfer source address register (DMSAR) and transfer destination address register (DMDAR).

The extended repeat area on the source address is specified by the DMAMD.SARA[4:0] bits. The extended repeat area on the destination address is specified by the DMAMD.DARA[4:0] bits. The size can be specified separately for the source and destination sides.

However, the area (of transfer source or transfer destination) which is specified as the repeat area or block area should not be specified as the extended repeat area.

When the address register value reaches the end address of the extended repeat area and the extended repeat area overflows, DMA transfer is stopped and an interrupt by an extended repeat area overflow can be requested. When an overflow occurs in the extended repeat area on the transfer source while the DMINT.SARIE bit is set to 1, the DMSTS.ESIF flag is set to 1 and the DMCNT.DTE bit is cleared to 0 to stop DMA transfer. At this time, if the DMINT.ESIE bit is set to 1, an interrupt by an extended repeat area overflow is requested. When the DMINT.DARIE bit is set to 1, the destination address register becomes a target to apply the function. DMA transfer can be resumed by writing 1 to the DMCNT.DTE bit in the interrupt handling.

Figure 17.7 shows an example of the extended repeat area operation.

Eight bytes are specified as an extended repeat area by the lower three bits of DMSAR (DMAMD.SARA[4:0] bits = 00011b). The data size is eight bits (DMTMD.SZ[1:0] = 00b).

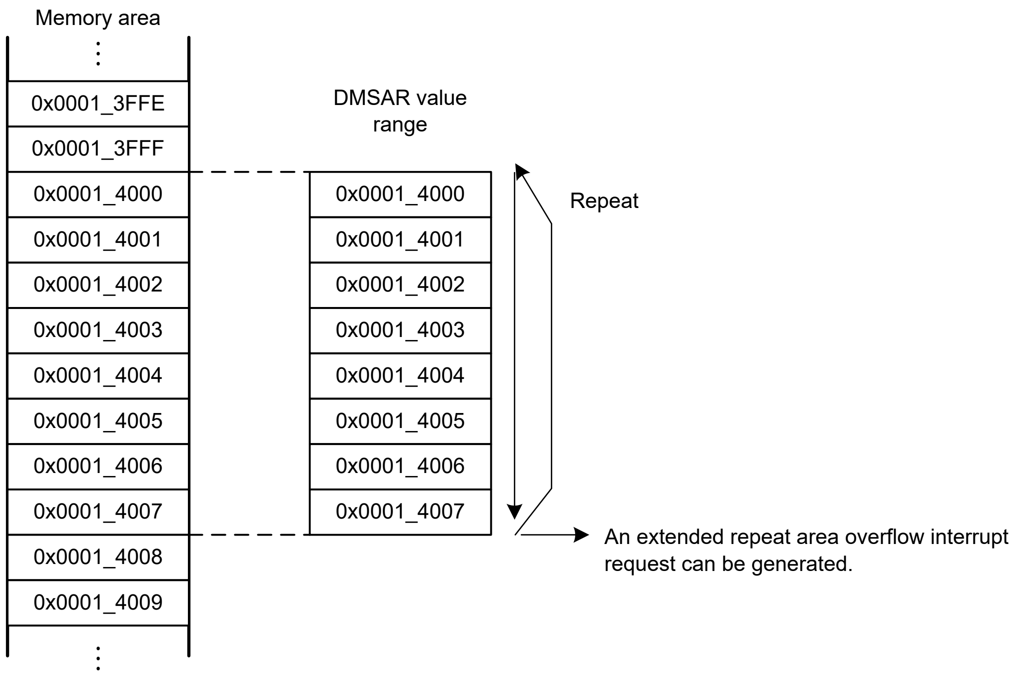

Figure 17.7 Example of extended repeat area operation

When an interrupt by an extended repeat area overflow is used in block transfer mode, the following should be taken into consideration.

When a transfer is stopped by an interrupt by an extended repeat area overflow, the address register must be set so that the block size is a power of 2 or the block size boundary is aligned with the extended repeat area boundary. When an overflow on the extended repeat area occurs during a transfer of one block, the interrupt by the overflow is suspended until transfer of the block is completed, and the transfer overruns.

Figure 17.8 shows an example when the extended repeat area function is used in block transfer mode.

Eight bytes are specified as an extended repeat area by the lower three bits of DMSAR (DMAMD.SARA[4:0] bits = 00011b), block transfer mode with block size 5 is set (DMCRA = 0x0005_0005), and transfer source address is not specified as a block area. Data size is eight bits (DMTMD.SZ[1:0] bits = 00b).

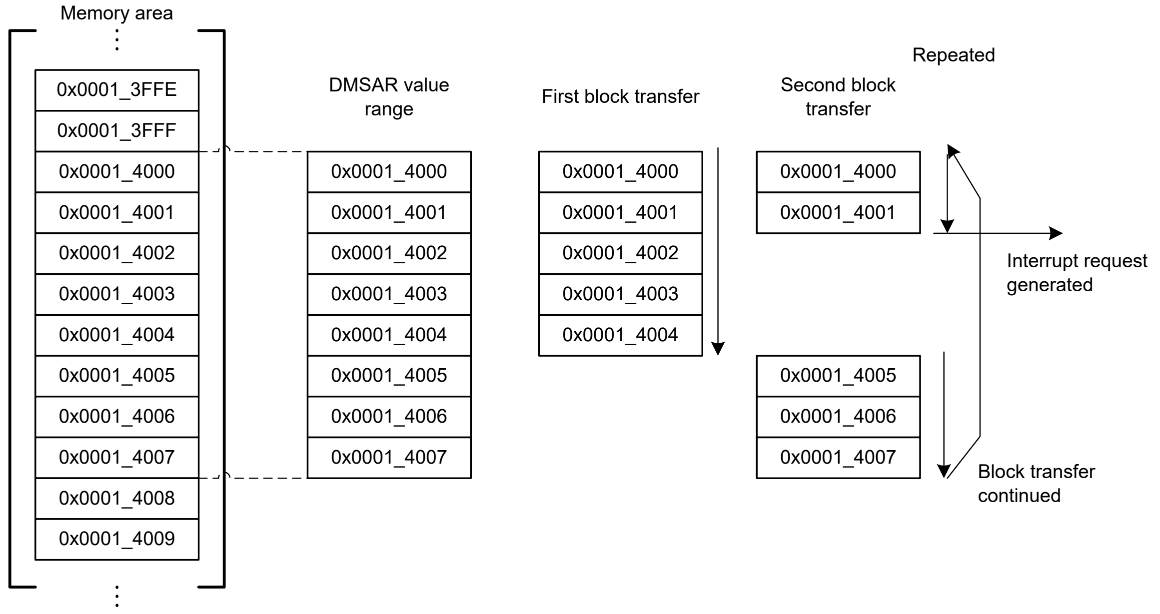

Figure 17.8 Example of extended repeat area function in block transfer mode

17.3.3 Free-running Function

The DMAC supports free-running function. This function allows to transfer repeatedly without reconfiguring in interrupt handler.

17.3.3.1 In Normal Transfer Mode

In normal transfer mode, when DMCRA.DMCRAL bits are set to 0x0000, no specific number of transfer operations is set; data transfer is performed with the transfer counter stopped.

For more information, see section 17.3.1.1. Normal Transfer Mode.

17.3.3.2 In Other Transfer Modes

In repeat, block and repeat-block transfer mode, the DMAC supports free-running function using the DMTMD.TKP bit. If the DMTMD.TKP bit is to be set to 1, the transfer is not stopped by completion of specified total number of transfer operations and reloads DMCRBH repeatedly.

Figure 17.9 show an example of block transfer operation without free-running function.

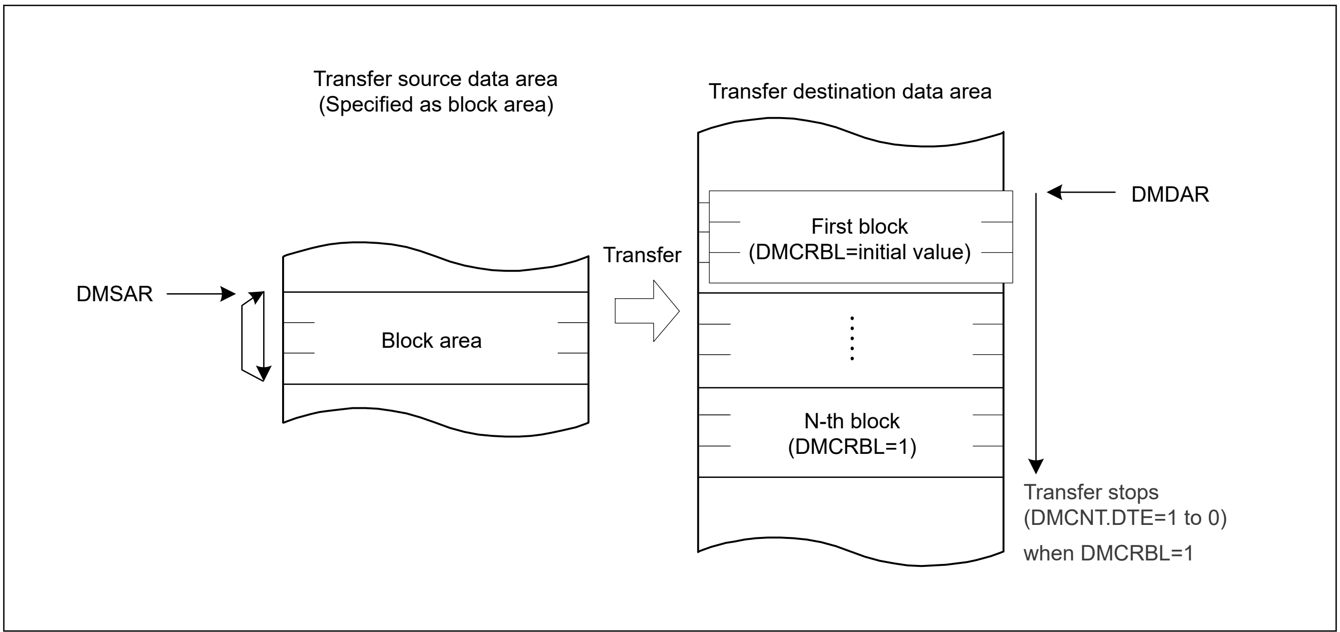

Figure 17.9 Operation in block transfer mode when DMTMD.TKP bit is set to 0

Figure 17.10 show an example of block transfer operation with free-running function.

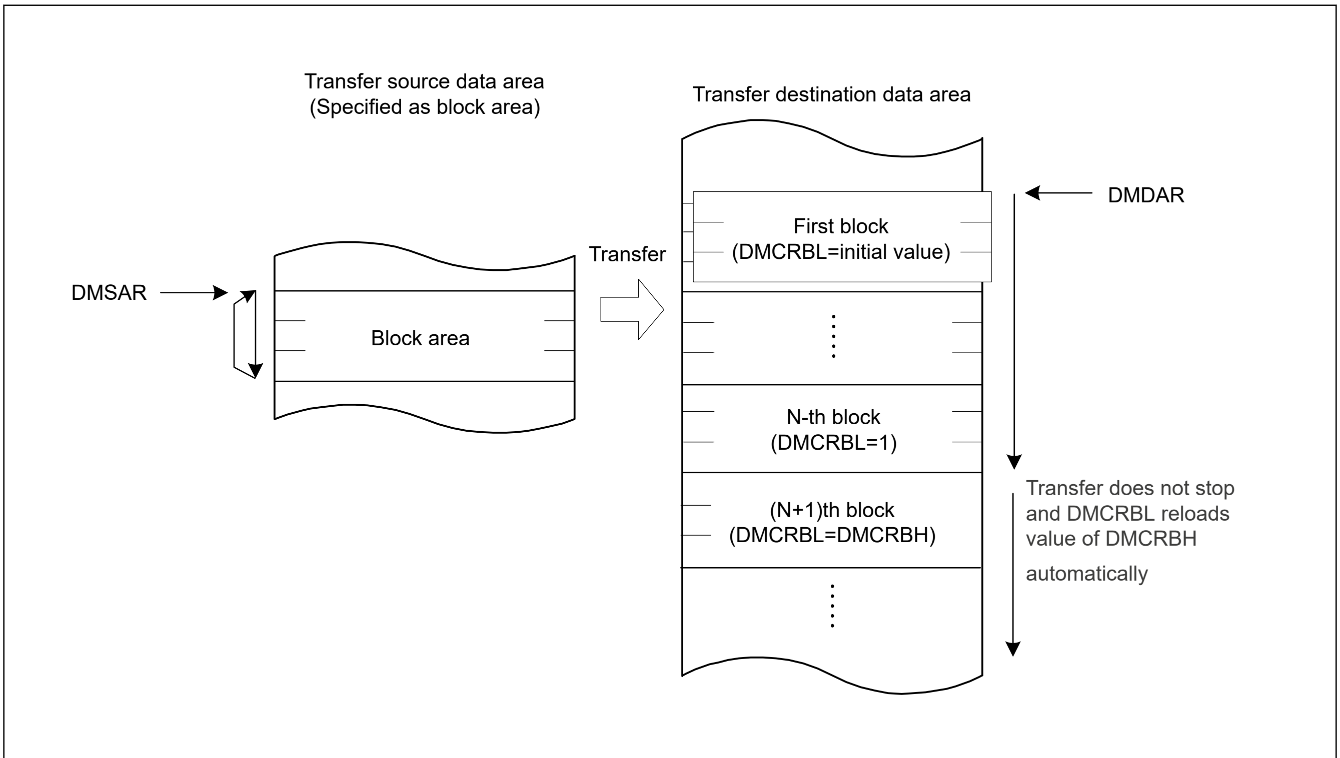

Figure 17.10 Operation in block transfer mode when DMTMD.TKP bit is set to 1

17.3.4 Address Update Function Using Offset

The source and destination addresses can be updated by fixing, increment, decrement, or offset addition. In normal, repeat and block transfer mode, when the offset addition is selected, the offset specified by the DMA offset register (DMOFR) is added to the address every time the DMAC performs one data transfer. This function realizes a data transfer where addresses are allocated to separated areas.

Offset subtraction can also be realized by setting a negative value in DMOFR. In this case, the negative value must be 2’s complement.

DMSBS or DMDBS are used instead of DMOFR in repeat-block transfer mode. For more information section 17.3.1.4. Repeat-Block Transfer Mode

Table 17.14 shows the address update method in each address update mode.

Table 17.14 Address update method in each address update mode

Address update mode	Settings of DMAMD.SM[1:0] and DMAMD.DM[1:0] for address update modes	Address update method (for different SZ[1:0] settings in DMTMD)			
		SZ[1:0] = 00b	SZ[1:0] = 01b	SZ[1:0] = 10b	SZ[1:0] = 11b
Address fixed	00b	Fixed			
Offset addition	01b	+DMOFR*1			
Increment	10b	+1	+2	+4	+8
Decrement	11b	–1	–2	–4	–8

Note 1. When setting a negative value in the DMA Offset Register, the value must be in two's complement, obtained by the following formula:
two's complement of a negative offset value = ~ (offset) + 1 (~ = bit inversion)

17.3.4.1 Basic Transfer Using Offset Addition

Figure 17.11 shows an example of address updating using offset addition.

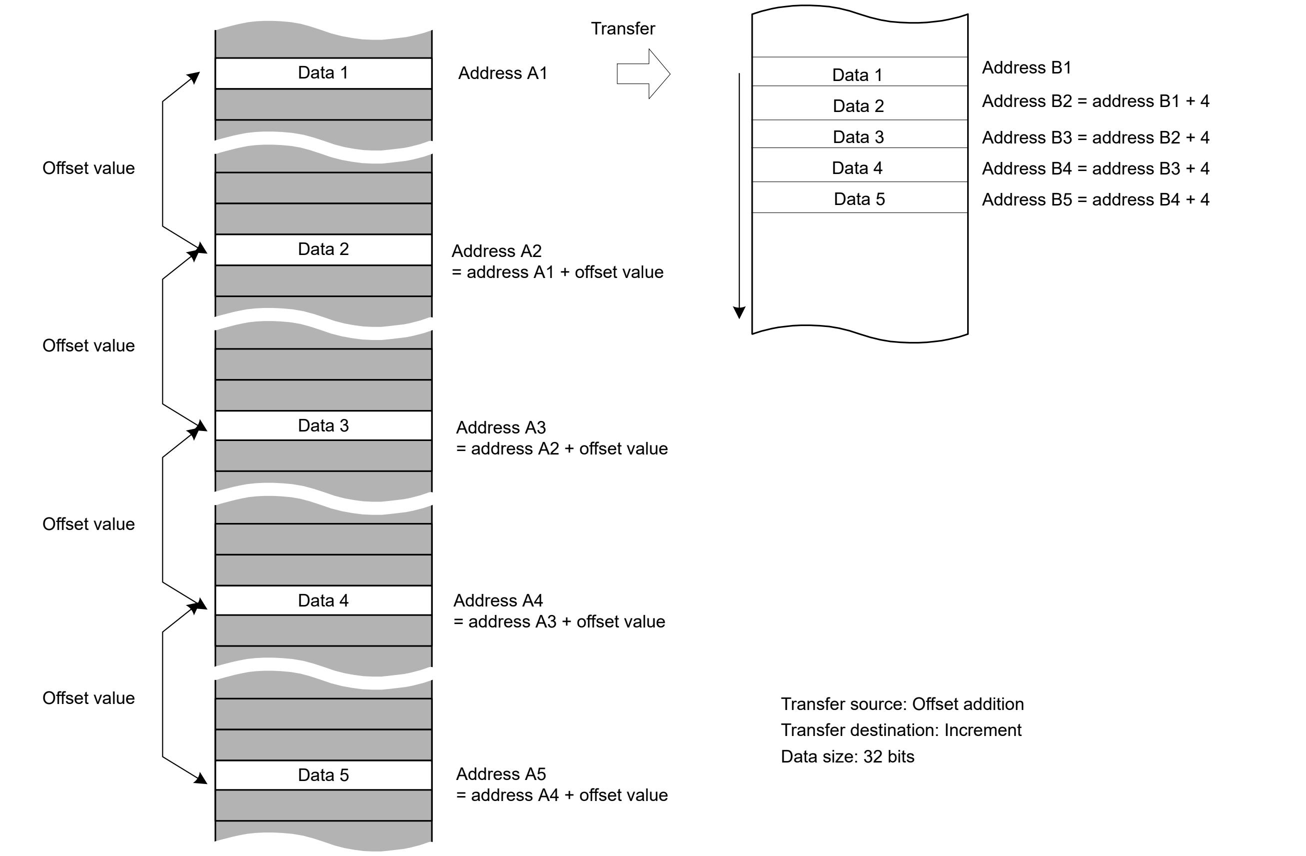

Figure 17.11 Example of address updating by offset addition

Figure 17.11 shows the setting of the following.

- The transfer data is 32 bits long.
- Offset addition is set as the transfer source address update mode.

- Increment is set as the transfer destination address update mode.

The second and subsequent data is each read from the transfer source address obtained by adding the offset value to the previous address. The data read from the addresses at the specified intervals is written to the continuous locations on the destination.

17.3.4.2 Example of XY Conversion Using Offset Addition

Figure 17.12 shows the XY conversion using offset addition in repeat transfer mode.

Settings are as follows:

- DMAMD.SM — Transfer source address update mode: Offset addition.
- DMAMD.DM — Transfer destination address update mode: Destination address is incremented.
- DMTMD.SZ — Transfer data size select: 32 bits.
- DMTMD.MD — Transfer mode select: Repeat transfer.
- DMTMD.DTS — Repeat area select: The source is specified as the repeat area.
- DMOFR — Offset address: 0x10.
- DMCRA — Repeat size: 0x4.
- DMINT.RPTIE — The repeat size end interrupt is enabled.

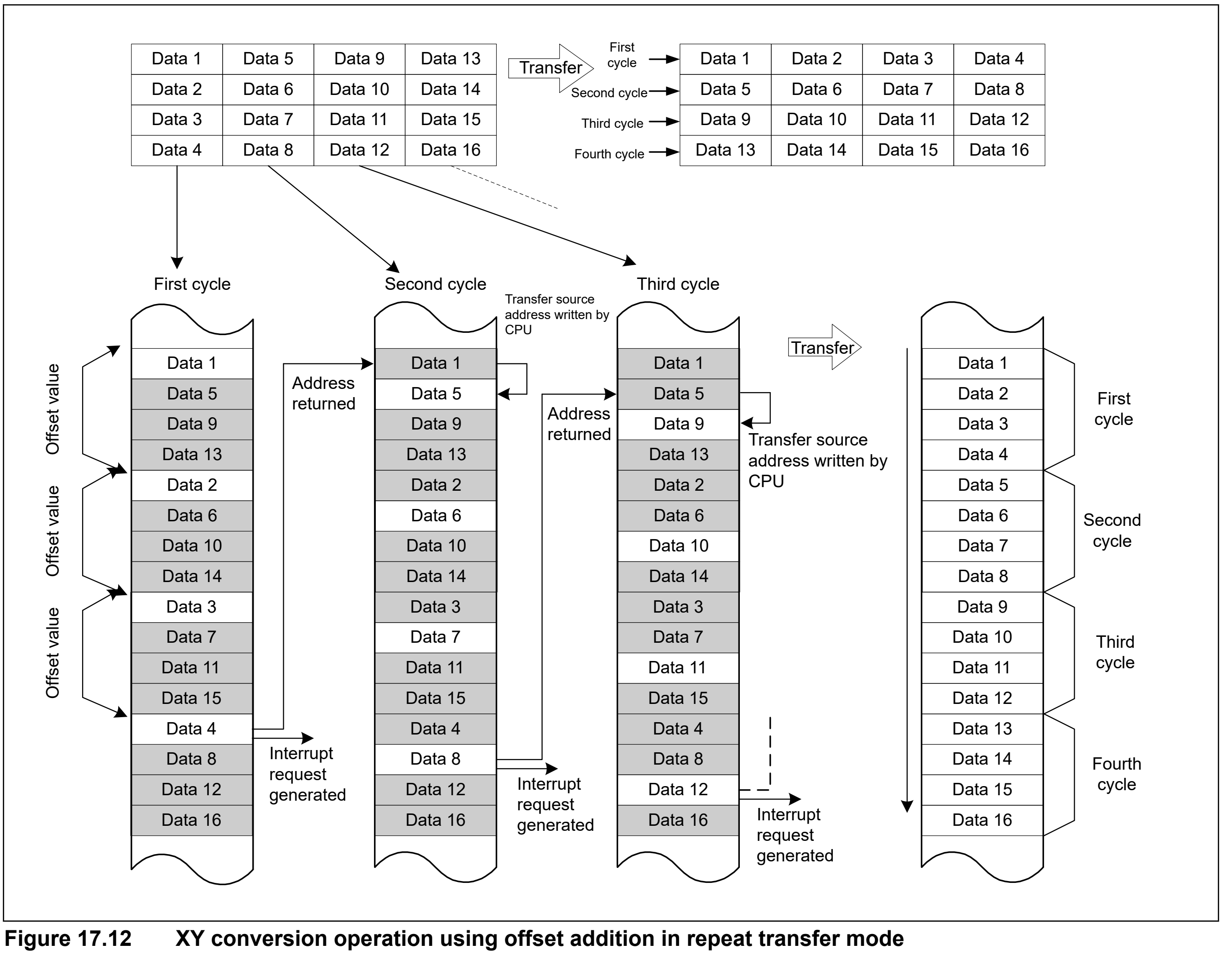

Figure 17.12 XY conversion operation using offset addition in repeat transfer mode

When a transfer starts, the offset value is added to the transfer source address every time data is transferred. The transfer data is written to continuous transfer destination addresses. When data 4 is transferred:

- The repeat size of data transfer is complete.
- The transfer source address returns to the transfer start address (the address of data 1 on the transfer source).
- A repeat size end interrupt is requested.

During the time this interrupt pauses the transfer, the following operations are performed.

- DMSAR — Rewrite the DMA transfer source address to the address of data 5 (with the above example, the data 1 address + 4).
- DMCNT — Set the DTE bit to 1.

The DMA transfer is resumed from the state when the DMA transfer is stopped. After that, the operations described above are repeated until the transfer source data is transposed to the destination area (XY conversion).

Figure 17.13 shows a flowchart of the XY conversion.

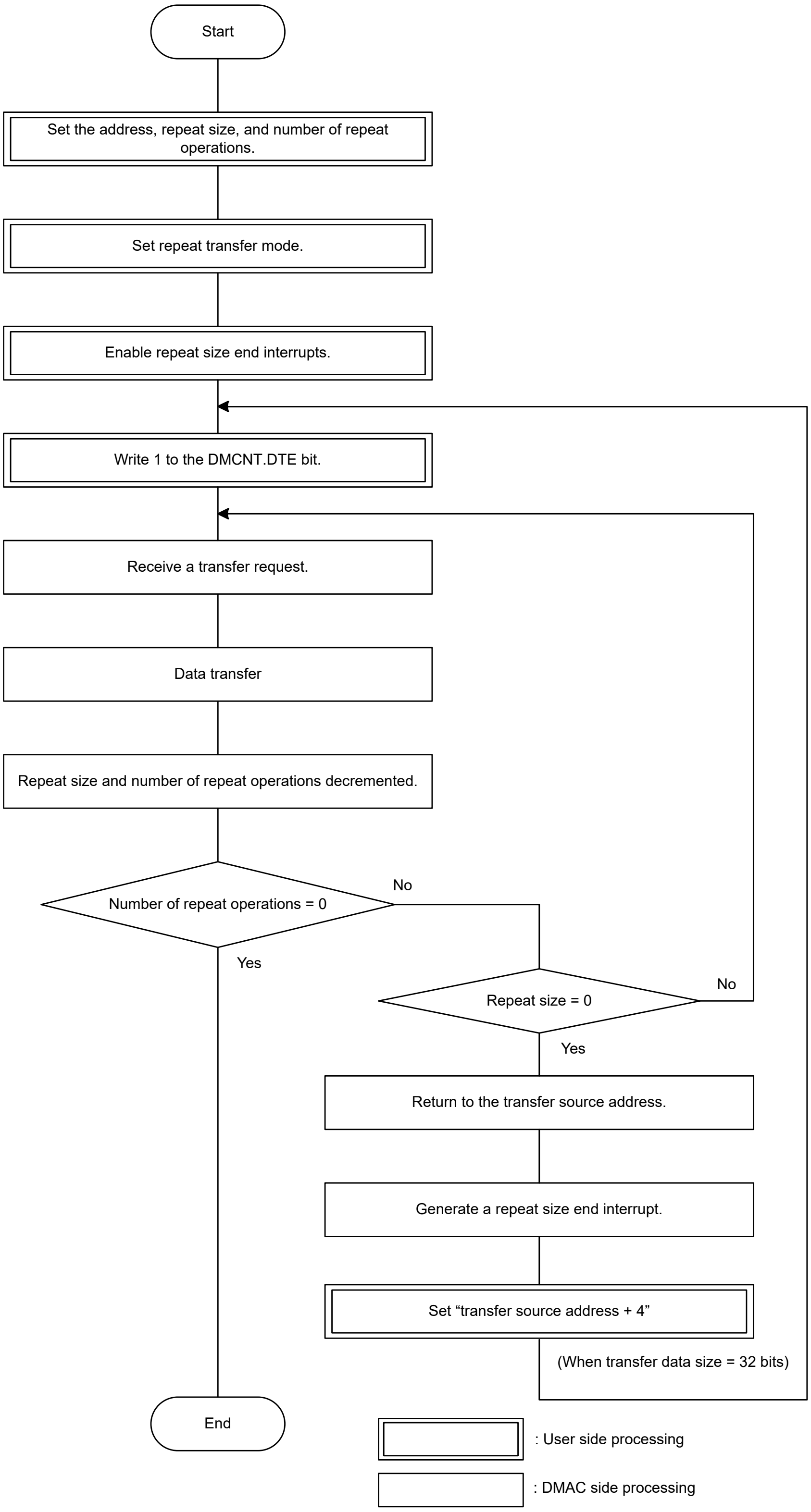

Figure 17.13 XY conversion flowchart using offset addition in repeat transfer mode

17.3.5 Address Update Function in Repeat-Block Transfer Mode

Repeat-block transfer mode is an extension of repeat transfer mode and block transfer mode. However, the detailed behavior of the address update is different from these two modes. Here are the details of the address update function in repeat-block transfer mode.

17.3.5.1 Fixed Address Mode

When DMAMD.SM[1:0] is set to 00b, the address update mode of the source is fixed address. And when DMAMD.DM[1:0] is set to 00b, the address update mode of the destination is fixed address.

In fixed address, the address is not updated from the initial value of DMSAR and DMDAR. If the block size (DMCRA) is larger than 1, the same data will be transferred multiple times for one request.

Figure 17.14 shows address update in fixed address.

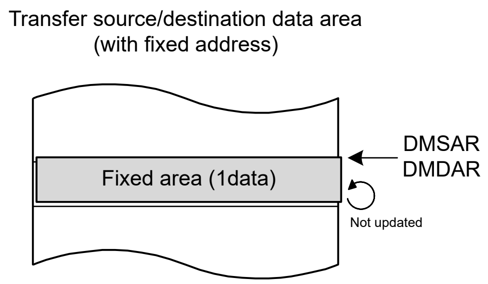

Figure 17.14 Address update in fixed address

17.3.5.2 Incremental and Decremental Address Mode

When DMAMD.SM[1:0] is set to 10b, the address update mode of the source is incremental address. And when DMAMD.DM[1:0] is set to 10b, the address update mode of the destination is incremental address. When DMAMD.SM[1:0] is set to 11b, the address update mode of the source is decremental address. And when DMAMD.DM[1:0] is set to 11b, the address update mode of the destination is decremental address.

In these update modes, the address is incremented or decremented according to the setting of DMTMD.SZ[1:0].

In these update modes DMSBS and DMDBS indicates a reload area. The unit of DMSBS and DMDBS is "number of data". At the start of transfer, DMSBSL and DMDBSL, which is the lower 16 bits of DMSBS and DMDBS, operates as a down counter and decrements each time one data transfer is performed. When the value becomes 1, DMSAR and DMDAR reloads the value of DMSRR and DMDRR.

Figure 17.15 shows address update in incremental address.

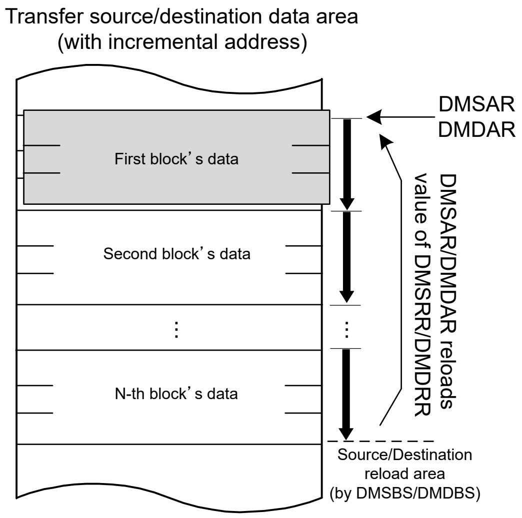

Figure 17.15 Address update in incremental address

17.3.5.3 Offset Addition Mode

When DMAMD.SM[1:0] is set to 01b, the address update mode of the source is offset addition. And when DMAMD.DM[1:0] is set to 01b, the address update mode of the destination is offset addition.

In offset addition, DMSBS and DMDBS indicates reload area and also works as an access offset value. Unlike other transfer modes, DMOFR register is not used in repeat-block transfer mode. In offset addition, the unit of DMSBS and DMDBS is the number of blocks. When the transfer starts, DMCRAL operates as a down counter, DMSAR and DMDAR reloads the value of DMSRR and DMDRR every time one block is transferred. In addition, DMSBSL and DMDBSL, which is the lower 16 bits of DMSBS and DMDBS, also operates as a down counter and decrements every time one block is transferred. When the DMSBS and DMDBS value becomes 1, DMSAR and DMDAR reloads the value of DMSRR and DMDRR.

When DMAMD.SADR and DMAMD.DADR is set to 0, offset addition operation of the same area is repeated. DMDAR only reloads DMDRR. Figure 17.16 shows address update in offset addition with DMAMD.SADR and DMAMD.DADR=0.

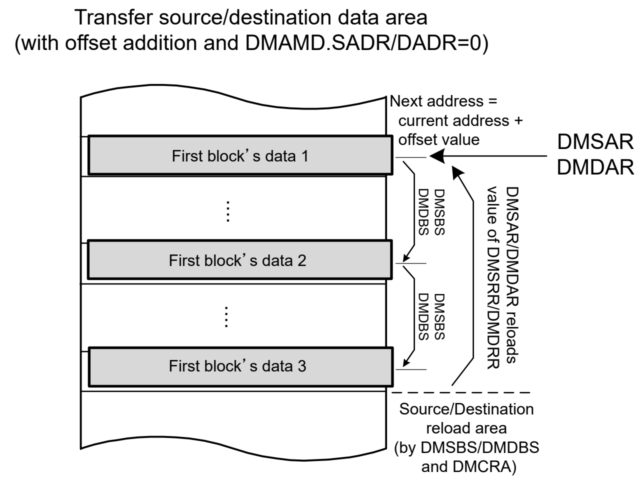

Figure 17.16 Address update in offset addition with DMAMD.SADR and DMAMD.DADR = 0

When DMAMD.SADR and DMAMD.DADR is set to 1, the address is incremented by one data unit after DMSRR and DMDRR is reloaded by DMCRAL=1. In other words, an index value ((DMDBSH-DMDBSL) × DataSize) is added to DMDAR after DMDRR is reloaded. This behavior is used to implement multiple ring buffers. Figure 17.17 shows address update in offset addition with DMAMD.SADR and DMAMD.DADR=1.

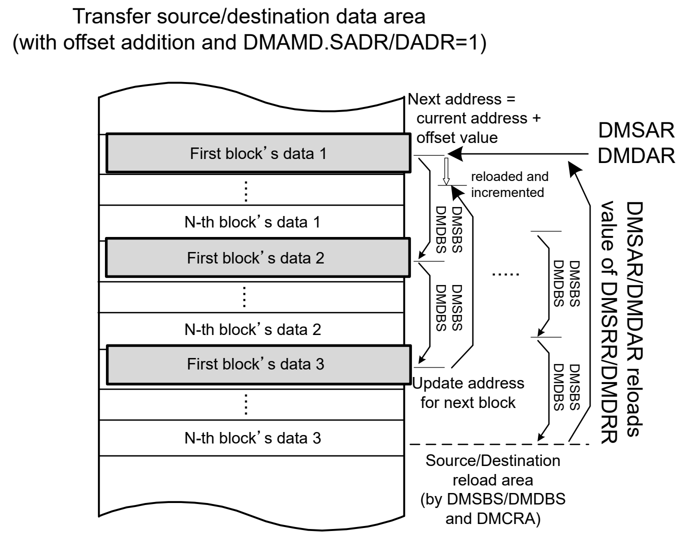

Figure 17.17 Address update in offset addition with DMAMD.SADR and DMAMD.DADR = 1

17.3.6 Example of Using Repeat-Block Transfer Mode

In repeat-block transfer mode, it is possible to realize repeated access to interval data and single or multiple ring buffers by combining the above address update modes. Following sections shows some usage examples.

17.3.6.1 Interval Address to Single Ring Buffer

Figure 17.18 shows an example of reading interval ADDRn registers (data register) of ADC16H module and storing it in single ring buffer. It transfers 2 data every 4 halfwords per 1 request. The block size (DMCRA) to 2, and the transfer source offset (DMSBS) to 4. Table 17.15 shows setting of this example.

Table 17.15 Setting of use case: from interval address to single ring buffer

Register	Value	Description
DMSAR, DMSRR	0x4033_A000	Initial source address
DMDAR, DMDRR	0x2200_0000	Initial destination address
DMTMD.SZ[1:0]	01b	Data size is halfword
DMAMD.SADR	0	DMSAR only reloads DMSRR
DMAMD.SM[1:0]	01b	Source update mode is offset addition
DMAMD.DM[1:0]	10b	Destination update mode is incremental address
DMCRAH, DMCRAL	2	Transfer block size
DMSBSH, DMSBSL	8	Source access offset (unit is ‘data’)
DMDBSH, DMDBSL	N × 2 (DMCRA)	Destination buffer size (unit is ‘data’)

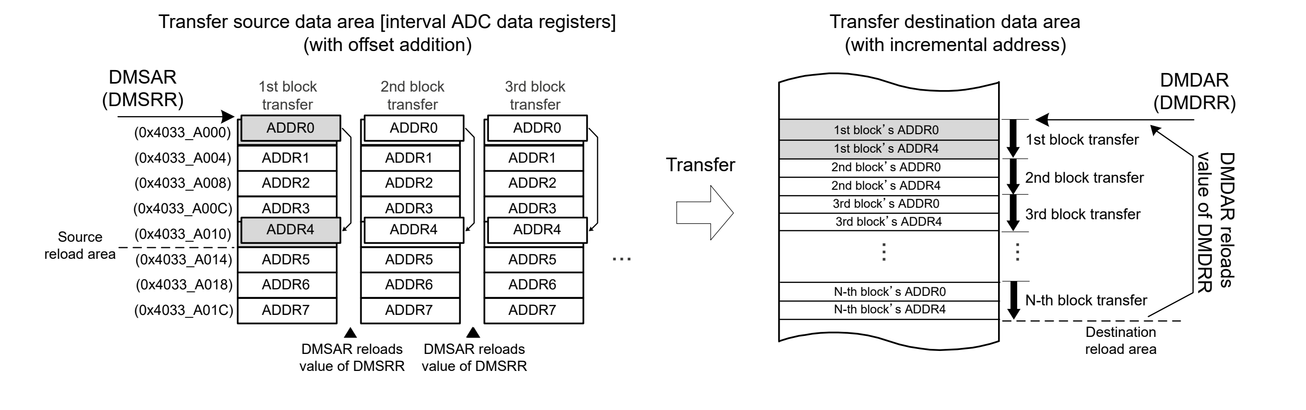

Figure 17.18 Example of use case: from interval address to single ring buffer

17.3.6.2 Single Block to Multi Ring Buffer

Figure 17.19 shows an example of storing the continuous ADDRn registers (data register) of ADC16H module individually in multiple ring buffers. In this example, a ring buffer in which only the first element (ADDR0) in a single block is arranged in transfer order is created at the destination. Also, in the next area, create a ring buffer in which only the second element (ADDR1) is arranged in transfer order. In the following case, create a ring buffer of length N, which is defined by DMDBS. And the number of data elements in the block is 3, which is defined by DMCRA. Table 17.16 shows setting of this example.

Table 17.16 Setting of use case: from single block to multi ring buffer (1 of 2)

Register	Value	Description
DMSAR, DMSRR	0x4033_A000	Initial source address
DMDAR, DMDRR	0x2200_0000	Initial destination address
DMTMD.SZ[1:0]	01b	Data size is halfword
DMAMD.DADR	1	Incremental destination address after reloading
DMAMD.SM[1:0]	01b	Source update mode is offset addition
DMAMD.DM[1:0]	01b	Destination update mode is offset addition

Table 17.16 Setting of use case: from single block to multi ring buffer (2 of 2)

Register	Value	Description
DMCRAH, DMCRAL	3	Transfer block size
DMSBSH, DMSBSL	2	Source buffer size (unit is ‘data’)
DMDBSH, DMDBSL	N	Destination whole buffer size (unit is ‘blocks’) and Destination access offset (unit is ‘data’)

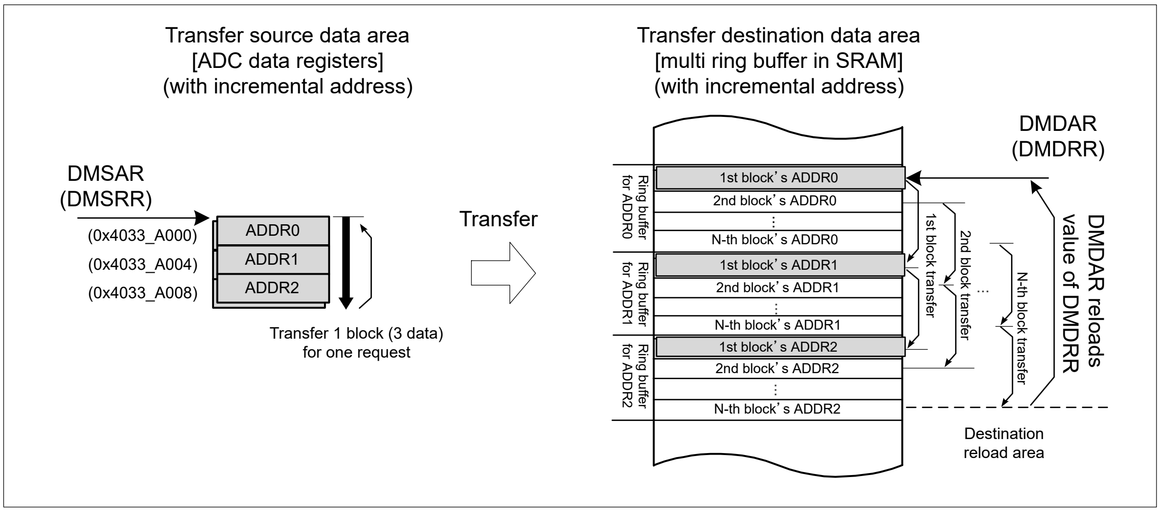

Figure 17.19 Example of use case: from single block to multi ring buffer

17.3.7 Activation Sources

Software, interrupt requests from the peripheral modules, and external interrupt requests can all be specified as DMAC activation sources. Set the DMTMD.DCTG[1:0] bits to select the activation source.

17.3.7.1 DMAC Activation by Software

When DMA transfer is started by software, follow below procedure.

1. Set the DMTMD.DCTG[1:0] bits to 00b.
2. Set the DMCNT.DTE bit to 1 (DMA transfer is enabled).
3. Set the DMAST.DMST bit set to 1 (DMAC activation enabled).
4. Set the DMREQ.SWREQ bit to 1 (DMA requested).

When the DMAC is activated by software while the DMREQ.CLRS bit is 0, the DMREQ.SWREQ bit is cleared to 0 after data transfer is started in response to a DMA transfer request.

When the DMAC is activated by software while the CLRS bit is 1, the SWREQ bit is not cleared to 0 after data transfer is started. In this case, a DMA transfer request is issued again after completion of a transfer.

17.3.7.2 DMAC Activation through Interrupt Requests from On-Chip Peripheral Modules or External Interrupt Requests

You can specify interrupt requests from on-chip peripheral modules and external interrupt requests as DMAC activation sources. The activation sources can be selected individually for each channel in the ICU.DELSRn.DELS[9:0] (n = 0 to 7) bits.

To start DMA transfer through an interrupt request from an on-chip peripheral module or an external interrupt request, follow the procedures as indicated below.

1. Set the ICU.DELSRn.DELS[9:0] (n = 0 to 7) bits to the event number (select the DMAC event link).

2. Set the DMTMD.DCTG[1:0] bits to 01b (interrupts from the peripheral modules and the external interrupt pins).
3. Set the DMCNT.DTE bit to 1 (enable DMA transfer).
4. Set the DMAST.DMST bit set to 1 (DMAC activation enabled)

For interrupt requests specified as DMAC activation sources, see Table 14.3 in section 14, Interrupt Controller Unit (ICU).

17.3.8 Operation Timing

The following timing charts show the minimum number of execution cycles.

Figure 17.20 and Figure 17.21 show DMAC operation timing examples.

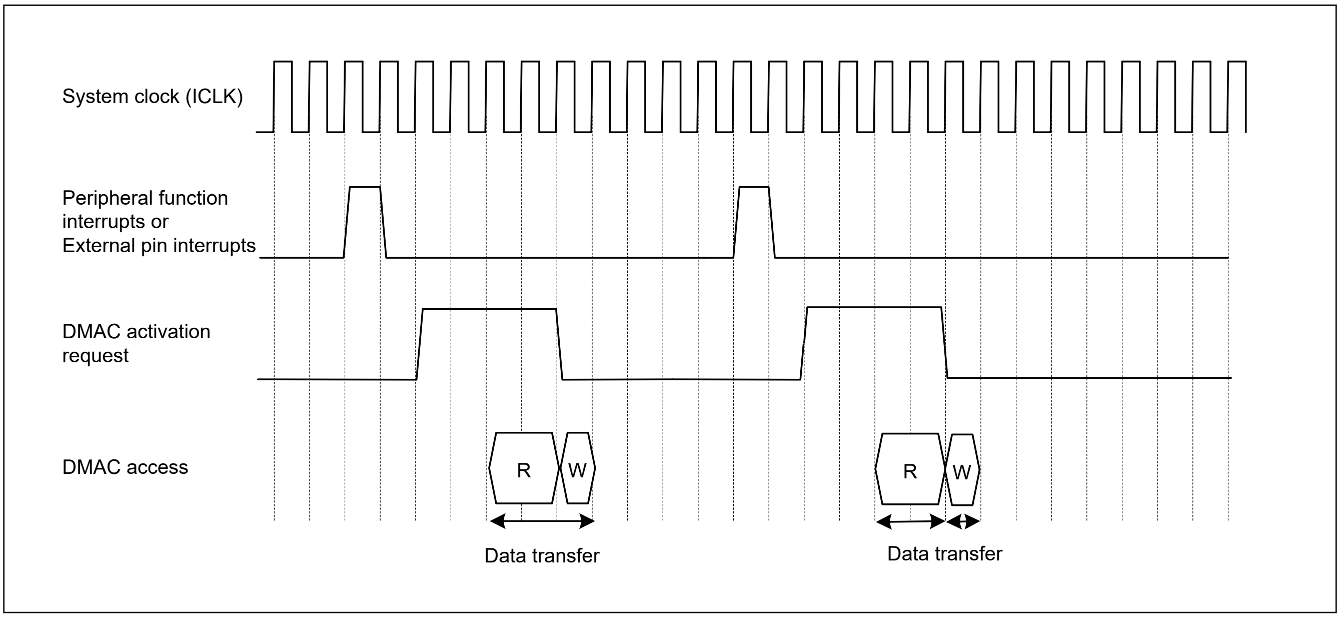

Figure 17.20 DMAC operation timing example 1 with DMAC activation by Interrupt from peripheral module or external interrupt input pin, in normal transfer mode or repeat transfer mode

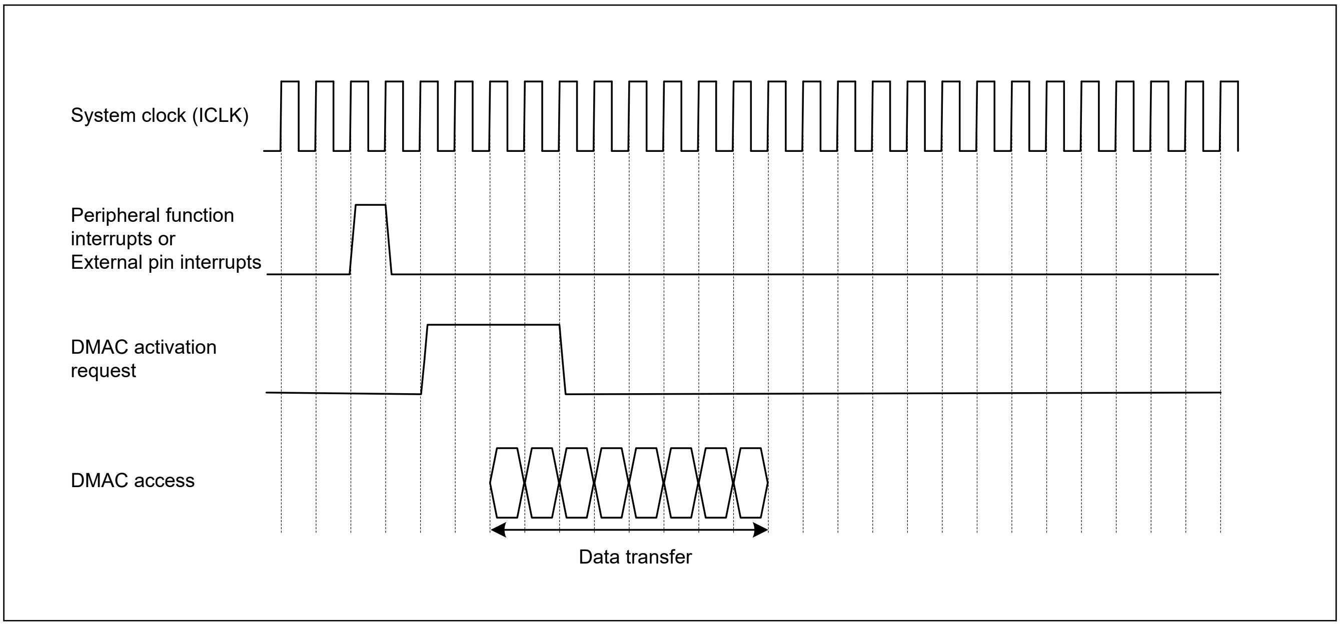

Figure 17.21 DMAC operation timing example 2 with DMAC activation by interrupt from peripheral module or external interrupt input pin, in block transfer mode with block size = 4

17.3.9 Activating the DMAC

Table 17.17 shows the register setting procedure of normal, repeat and block transfer mode and Table 17.18 shows register setting procedure of repeat-block transfer mode.

Table 17.17 Register Setting Procedure of Normal Transfer Mode, Repeat Transfer Mode and Block Transfer Mode

No.	Step Name	Description
1	Disable the peripheral function as the DMAC request source.	To use peripheral function interrupts as DMAC activation sources. Disable the control register for the peripheral function.
2	Disable the IRQn pin as the DMAC request source.	To use external pin interrupts as DMAC activation sources.
3	Set the ICU.DELSRn.DELS[9:0] bits to 0x00.	Disable the DMAC request.
4	Clear the DMCNT.DTE bit to 0	Disable DMA transfer.
5	Set the interrupt request as a DMAC request source in the DELSRn register by using the ICU.	To use internal peripheral interrupts or external pin interrupts as DMAC activation sources. Enable the interrupt bit for the activation source. Set the DMAC activation source.
6	Set the peripheral module as a DMAC request source	To use peripheral function interrupt as a DMAC activation source. Set the control register for the peripheral function without starting it.
7	Set the IRQn pin function by using the ICU.	To use external pin interrupt as a DMAC activation source. Set the IRQn pin function by using the Interrupt Controller Unit.
8	Set the DMAMD.DM[1:0] bits Set the DMAMD.SM[1:0] bits Set the DMAMD.DARA[4:0] bits Set the DMAMD.SARA[4:0] bits	Set the Transfer destination address update mode bits Set the Transfer source address update mode bits Set the Transfer destination address extended repeat area bits Set the Transfer source address extended repeat area bits
9	Set the DMTMD.DCTG[1:0] bits Set the DMTMD.SZ[1:0] bits Set the DMTMD.DTS[1:0] bits Set the DMTMD.MD[1:0] bits Set the DMTMD.TKP bit	Set the Transfer request select bits Set the Data transfer size bits Set the Repeat area select bits Set the Transfer mode select bits Set the transfer keeping select bit
10	Set the DMSAR register Set the DMDAR register Set the DMCRA register	Set the transfer source start address. Set the transfer destination start address. Set the number of transfer operations.
11	Set the DMCRB register	To use block transfer mode or repeat transfer mode. Set the number of block transfer operations.
12	Set the DMOFR register	To use the address update function with offset. Set the offset value.
13	Set the DMINT.DTIE bit to 1	To use the DMA transfer end interrupts. Enable DMAC transfer end interrupts.
14	Set the DMINT.RPTIE bit Set the DMINT.SARIE bit Set the DMINT.DARIE bit Set the DMINT.ESIE bit to 1	To use the DMA transfer escape end interrupts Set the repeat size end interrupt. Set the transfer source address extended repeat area overflow interrupt. Set the transfer destination address extended repeat area overflow interrupt. Enable the DMA transfer escape end interrupt.
15	Set the DMCNT.DTE bit to 1	Enable DMA transfer.
16	Set the DMAST.DMST bit to 1	Enable DMAC operation. *1 Common settings for DMAC
17	Start the peripheral function as a DMAC request source	To use peripheral function interrupt as a DMAC activation source
18	Enable the IRQn pin as a DMAC request source	To use external pin interrupt as a DMAC activation source
19	End of initial settings	For activation by software On completion of the initial settings, writing 1 to the DMA software start bit (DMREQ.SWREQ) starts DMA transfer.

Note 1. The DMAST.DMST bit setting does not necessarily have to follow the settings for the individual activation sources.

Table 17.18 Register Setting Procedure of Repeat-Block Transfer Mode

No.	Step Name	Description
1	Disable the peripheral function as the DMAC request source.	To use peripheral function interrupts as DMA activation sources. Disable the control register for the peripheral function.
2	Disable the IRQ pin as the DMAC request source.	To use external pin interrupts as DMA activation sources.
3	Set the ICU.DELSRn.DELS[9:0] bits to 0x00.	Disable the DMAC request.
4	Clear the DMCNT.DTE bit to 0	Disable DMAC transfer.
5	Set the interrupt request as a DMAC request source in the DELSRn register by using the ICU.	To use internal peripheral interrupts or external pin interrupts as DMA activation sources. Enable the interrupt bit for the activation source. Set the DMAC activation source.
6	Set the peripheral module as a DMAC request source	To use peripheral function interrupt as a DMA activation source. Set the control register for the peripheral function without starting it.
7	Set the IRQ pin function by using the Interrupt Controller Unit.	To use external pin interrupt as a DMA activation source. Set the IRQ pin function by using the Interrupt Controller Unit.
8	Set the DMAMD.DM[1:0] bits Set the DMAMD.SM[1:0] bits Set the DMAMD.DADR bit Set the DMAMD.SADR bit	Set the Transfer destination address update mode bits Set the Transfer source address update mode bits Set the Transfer destination address update select after reload Set the Transfer source address update select after reload
9	Set the DMTMD.DCTG[1:0] bits Set the DMTMD.SZ[1:0] bits Set the DMTMD.MD[1:0] bits Set the DMTMD.TKP bit	Set the Transfer request select bits Set the Data transfer size bits Set the Transfer mode to repeat-block transfer mode Set the transfer keeping select bit
10	Set the DMSAR register Set the DMDAR register Set the DMSRR register Set the DMDRR register Set the DMCRA register Set the DMCRB register	Set the transfer source start address Set the transfer destination start address Set the initial value of source start address Set the initial value of destination start address Set the number of transfer operations Set the number of block transfer operations
11	Set the DMSBS register Set the DMDBS register	To use the address update function with incremental, decremental or offset Set the source buffer size and access offset Set the destination buffer size and access offset
12	Set the DMINT.DTIE bit to 1	To use DMA transfer end interrupts. Enable DMAC transfer end interrupts.
13	Set the DMCNT.DTE bit to 1	Enable DMAC transfer
14	Set the DMAST.DMST bit to 1	Enable DMAC operation. *1
15	Start the peripheral function as a DMAC request source	To use peripheral function interrupt as a DMA activation source
16	Enable the IRQ pin as a DMAC request source	To use external pin interrupt as a DMA activation source
17	End of initial settings	For activation by software On completion of the initial settings, writing 1 to the DMA software start bit (DMREQ.SWREQ) starts DMA transfer.

Note 1. The DMAST.DMST bit setting does not necessarily have to follow the settings for the individual activation sources.

17.3.10 Starting DMA Transfer

To enable the DMA transfer, set the DMCNT.DTE bit to 1 (enable the DMA transfer), and then set the DMAST.DMST bit to 1 (enable the DMAC activation).

New activation requests are not accepted during the transfer of another DMAC channel or DTC. When the preceding transfer is complete, channel arbitration selects the DMA transfer request of the highest priority channel, and the DMA transfer of that channel starts. When the DMA transfer starts, the DMSTS.ACT flag is set to 1 (the DMAC is in the active state).

17.3.11 Registers During DMA Transfer

The DMAC registers are updated by a DMA transfer. The value to be updated differs according to the other settings and the transfer state. The registers to be updated are DMSAR, DMDAR, DMCRA, DMCRB, DMSBS, DMDBS, DMCNT, and DMSTS.

DMA Source Address Register (DMSAR)

When data has been transferred in response to one transfer request, the contents of DMSAR are updated to the address to be accessed by the next transfer request.

For details on register update operation in each transfer mode, see Table 17.5 to Table 17.13.

DMA Destination Address Register (DMDAR)

When data has been transferred in response to one transfer request, the contents of DMDAR are updated to the address to be accessed by the next transfer request.

For details on register update operation in each transfer mode, see Table 17.5 to Table 17.13.

DMA Transfer Count Register (DMCRA)

When data has been transferred in response to one transfer request, the count value is updated. The update operation depends on the transfer mode selected.

For details on register update operation in each transfer mode, see Table 17.5 to Table 17.13.

DMA Block Transfer Count Register (DMCRB)

When data has been transferred in response to one transfer request, the count value is updated. The update operation depends on the transfer mode selected.

For details on register update operation in each transfer mode, see Table 17.5 to Table 17.13.

DMA Source Buffer Size Register (DMSBS)

When data has been transferred in response to one transfer request, the count value is updated. The update operation depends on the transfer mode selected.

For details on register update operation in each transfer mode, see Table 17.8 to Table 17.13.

DMA Destination Buffer Size Register (DMDBS)

When data has been transferred in response to one transfer request, the count value is updated. The update operation depends on the transfer mode selected.

For details on register update operation in each transfer mode, see Table 17.8 to Table 17.13.

DMA Transfer Enable Bit (DMCNT.DTE)

Although the DMCNT.DTE bit enables or disables data transfer by the register write access, it is automatically cleared to 0 by the DMAC according to the DMA transfer state.

The conditions for clearing this bit by the DMAC are as follows:

- When the specified total volume of data transfer is completed
- When DMA transfer is stopped by the repeat size end interrupt
- When DMA transfer is stopped by the extended repeat area overflow interrupt
- When DMA transfer error occurs

Writing to the registers for the channels when the corresponding DMCNT.DTE bit is set to 1 is prohibited (except for DMCNT). In this case, writing must be performed after the bit is cleared to 0.

DMAC Active Flag (DMSTS.ACT)

The DMSTS.ACT flag indicates whether the DMACn is in the idle or active state.

This flag is set to 1 when the DMAC starts data transfer, and is cleared to 0 when data transfer in response to one transfer request is completed.

Even when DMA transfer is stopped by writing 0 to the DMCNT.DTE bit during DMA transfer, this flag remains 1 until DMA transfer is completed.

Transfer End Interrupt Flag (DMSTS.DTIF)

The DMSTS.DTIF flag is set to 1 after DMA transfer of the total transfer size of data is completed.

When both this flag and the DMINT.DTIE bit are set to 1, a transfer end interrupt is requested.

This flag is set to 1 when the DMA transfer bus cycle is completed and the DMSTS.ACT flag is cleared to 0 indicating the DMA transfer end.

This flag is automatically cleared to 0 when the DMCNT.DTE bit is set to 1 during the interrupt handling.

Transfer Escape End Interrupt Flag (DMSTS.ESIF)

The DMSTS.ESIF flag is set to 1 when a repeat size end interrupt or extended repeat area overflow interrupt is requested. When this bit and the DMINT.ESIE bit are set to 1, a transfer escape end interrupt is requested.

This flag is set to 1 when the bus cycle of the DMA transfer having caused the interrupt request is completed and the DMSTS.ACT flag is cleared to 0 indicating the DMA transfer end.

This flag is automatically cleared to 0 when the DMCNT.DTE bit is set to 1 during an interrupt handling.

Before sending an interrupt request from the DMAC to the CPU or the DTC, the interrupt control register must be set.

For details, see section 14, Interrupt Controller Unit (ICU).

17.3.12 Channel Priority

When multiple DMA transfer requests and a DTC transfer request are present, the DMAC and DTC determine the priority of channels that have DMA transfer requests and a DTC transfer request.

- When DMCTL.PR = 0, the channel priority is fixed as follows: channel 0 > channel 1 > channel 2 > channel 3 … > channel 7 > DTC (channel 0: Highest).

When DMCTL.PR = 1, the DMAC and DTC determine the priority of channels at round-robin.

When a DMA transfer request is generated during data transfer, channel arbitration is started after the final data has been transferred, and DMA transfer of the higher-priority channel starts.

17.3.13 Channel Security

The secure attribute can be set with CPSCU.DMACCHSAR.SADMACm0n (m = 0, 1, n = 0 to 7) for each DMAC channel.

When the CPSCU.DMACCHSAR.SADMACm0n bit is 0.

- When the corresponding channel transfers, it behaves as a secure master.
- The register of corresponding channel has a secure attribute.

When the CPSCU.DMACCHSAR.SADMACm0n bit is 1.

- When the corresponding channel transfers, it behaves as a non-secure master.
- The register of corresponding channel has a non-secure attribute.

Refer to the Security chapter and BUS chapter for areas accessible to secure and non-secure masters.

Do not change to the CPSCU.DMACCHSAR while DMA transfer.

17.3.14 Channel Privilege

The privileged attribute can be set with CPSCU.DMACCHPAR.PADMACm0n (m = 0, 1, n = 0 to 7) for each DMAC channel n.

When the CPSCU.DMACCHPAR.PADMACm0n bit is 0.

- When the corresponding channel transfers, it behaves as a privileged master.
- The registers of channel are protected from an unprivileged access.

When the CPSCU.DMACCHPAR.PADMACm0n bit is 1.

- The transfer of DMAC channel is unprivileged access for both read and write.
- The registers of channel are unprivileged attributes.

Do not change to the CPSCU.DMACCHPAR while DMA transfer.

17.4 Ending DMA Transfer

The operation for ending DMA transfer depends on the transfer end conditions. When DMA transfer ends, the DMCNT.DTE bit and the DMSTS.ACT flag are changed from 1 to 0, indicating that DMA transfer has ended.

Before sending an interrupt request from the DMAC to the CPU or the DTC, the interrupt control register must be set.

For details, see section 14, Interrupt Controller Unit (ICU).

17.4.1 Transfer End by Completion of Specified Total Number of Transfer Operations

(1) In Normal Transfer Mode (DMTMD.MD[1:0] = 00b)

When the value of DMCRAL changes from 1 to 0, DMA transfer ends on the corresponding channel, and the DMCNT.DTE bit is cleared to 0 and the DMSTS.DTIF flag is set to 1 at the same time. If the DMINT.DTIE bit is 1 at this time, a transfer end interrupt request is issued to the CPU or the DTC.

(2) In Repeat Transfer Mode (DMTMD.MD[1:0] = 01b)

When the value of DMCRBL changes from 1 to 0, DMA transfer ends on the corresponding channel, and the DMCNT.DTE bit is cleared to 0 and the DMSTS.DTIF flag is set to 1 at the same time. If the DMINT.DTIE bit is 1 at this time, an interrupt request is issued to the CPU or the DTC.

If the DMTMD.TKP bit is 1 (in free-running function), the DMSTS.DTIF bit is set to 1, but the DMCNT.DTE bit is not cleared to 0.

(3) In Block Transfer Mode (DMTMD.MD[1:0] = 10b)

When the value of DMCRBL changes from 1 to 0, DMA transfer ends on the corresponding channel, and the DMCNT.DTE bit is cleared to 0 and the DMSTS.DTIF flag is set to 1 at the same time. If the DMINT.DTIE bit is 1 at this time, an interrupt request is issued to the CPU or the DTC.

If the DMTMD.TKP bit is 1 (in free-running function), the DMSTS.DTIF bit is set to 1, but the DMCNT.DTE bit is not cleared to 0.

(4) In Repeat-Block Transfer Mode (DMTMD.MD[1:0] = 11b)

When the value of DMCRBL changes from 1 to 0, DMA transfer ends on the corresponding channel, and the DMCNT.DTE bit is cleared to 0 and the DMSTS.DTIF flag is set to 1 at the same time. If the DMINT.DTIE bit is 1 at this time, an interrupt request is issued to the CPU or the DTC.

If the DMTMD.TKP bit is 1 (in free-running function), the DMSTS.DTIF bit is set to 1, but the DMCNT.DTE bit is not cleared to 0.

17.4.2 Transfer End by Repeat Size End Interrupt

In repeat transfer mode, a repeat size end interrupt is requested when transfer of a 1-repeat size of data is completed while the DMINT.RPTIE bit is set to 1. When the interrupt is requested to complete DMA transfer, the DMCNT.DTE bit is cleared to 0 and the DMSTS.ESIF flag is set to 1 even if the DMTMD.TKP bit is 1 (in free-running function). If the DMINT.ESIE bit is 1 at this time, an interrupt request is issued to the CPU or the DTC. Here, the transfer can be resumed by writing 1 to the DMCNT.DTE bit.

A repeat size end interrupt can be requested also in block transfer mode. In block transfer mode, the interrupt is requested in the same way as in repeat transfer mode when transfer of a 1-block size data is completed.

Repeat size end interrupt cannot be requested in repeat-block transfer mode.

Before sending an interrupt request from the DMAC to the CPU or the DTC, the interrupt control register must be set. For details, see section 14, Interrupt Controller Unit (ICU).

17.4.3 Transfer End by Interrupt on Extended Repeat Area Overflow

When an overflow on the extended repeat area occurs while the extended repeat area is specified and the DMINT.SARIE or DMINT.DARIE bit is set to 1 even if the DMTMD.TKP bit is 1 (in free-running function), an interrupt by an extended repeat area overflow is requested. When the interrupt is requested, the DMA transfer is terminated, the DMCNT.DTE bit is cleared to 0, and the ESIF flag in DMSTS is set to 1. If the DMINT.ESIE bit is 1 at this time, an interrupt request is issued to the CPU or the DTC.

Even if an interrupt by an extended repeat area overflow is requested during a read cycle, the following write cycle is performed.

In block transfer mode, even if an interrupt by an extended repeat area overflow is requested during a 1-block transfer, the remaining data in the block is transferred; transfer is terminated after a block transfer.

An interrupt by an extended repeat area overflow cannot be requested in repeat-block transfer mode.

Before sending an interrupt request from the DMAC to the CPU or the DTC, the interrupt control register must be set. For details, see section 14, Interrupt Controller Unit (ICU).

17.5 Processing on DMA Transfer Error

DMA transfer error occurs with MSAU error, the Slave TrustZone Filter error, the Master MPU error, the Slave Bus Error or the Illegal Access Error. If the access error occurs during the DMA transfer, the DMAC immediately stops the transfer of error occurred channel. If there is a request other than the channel which caused the error, it will be re-arbitration as it is.

When the transfer error occurs, DMCNT.DTE of the error causing channel is set to 0. Also, ICU.DELSRn of the corresponding channel is cleared. Write back to each register is not performed. The information of the channel that caused the error is set in DMECHR.

17.6 Interrupts

17.6.1 Transfer End Interrupt

Each DMAC channel can output an interrupt request DMACmn_INT (m = 0, 1, n = 0 to 7) to the CPU or the DTC after transfer in response to one request is completed.

The interrupt request is generated after the bus response to the final write of the DMA transfer. The timing of the bus response to a DMA write access depends on the DMBWR.BWE and transfer destination. For details, see section 17.9.1. Bus Response to DMA Write Access.

In repeat-block transfer mode, do not enable escape transfer end interrupt.

Table 17.19 lists the relation among the interrupt sources, the interrupt status flags, and the interrupt enable bits. Figure 17.22 shows the schematic logic diagram of interrupt outputs (DMACn (n = 0 to 7)). Figure 17.23 shows the DMAC interrupt handling routine to resume/terminate DMA transfer.

Table 17.19 Relation among interrupt sources, interrupt status flags, and interrupt enable bits

Interrupt sources		Interrupt enable bits	Interrupt status flags	Request output enable bits
Transfer end		—	DMSTS.DTIF	DMINT.DTIE
Escape transfer end	Repeat size end	DMINT.RPTIE	DMSTS.ESIF	DMINT.ESIE
	Source address extended repeat area overflow	DMINT.SARIE		
	Destination address extended repeat area overflow	DMINT.DARIE		

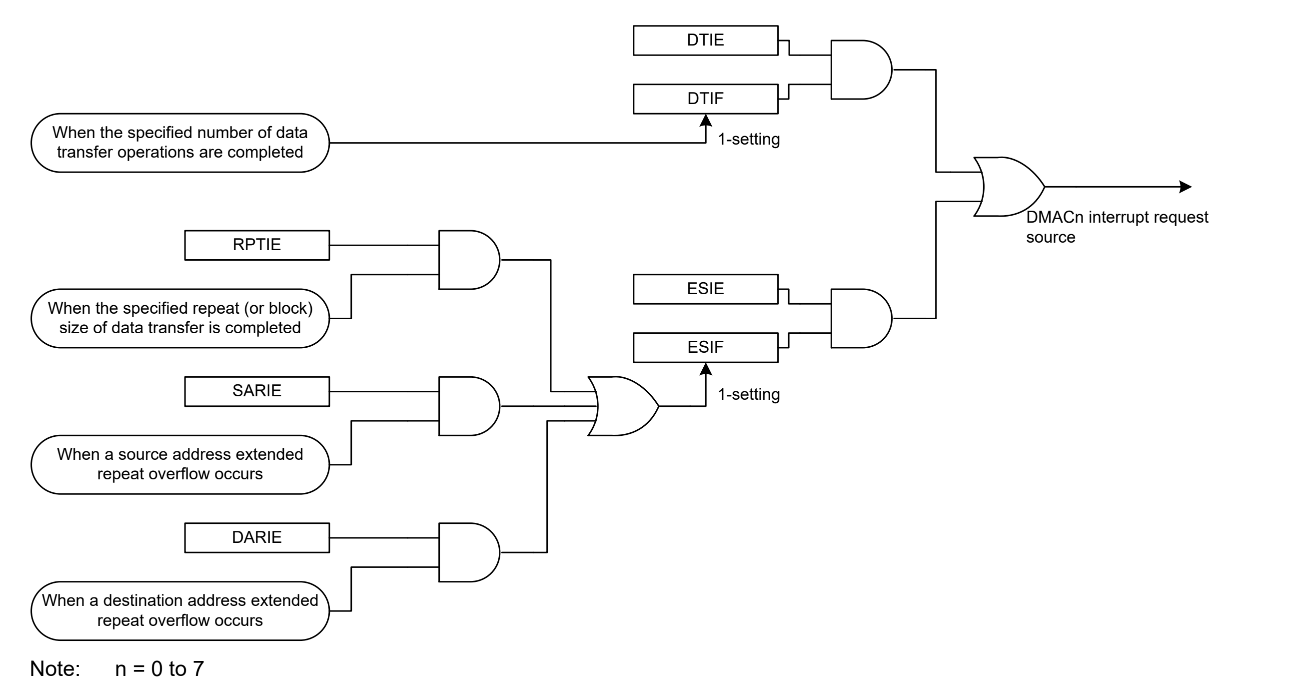

Figure 17.22 Schematic logic diagram of interrupt output source (DMACn)

Specifically, the different procedures are used for canceling an interrupt to restart DMA transfer in the following two cases:

- When terminating a DMA transfer
- When continuing a DMA transfer

17.6.1.1 When Terminating a DMA Transfer

Write 0 to the DMSTS.DTIF flag to clear a transfer end interrupt, and to the DMSTS.ESIF flag to clear a repeat size interrupt and an extended repeat area overflow interrupt. The DMACn remains in the stop state. When starting another DMA transfer after that, set the appropriate registers, and set the DMCNT.DTE bit to 1 (DMA transfer enabled).

17.6.1.2 When Continuing a DMA Transfer

Write 1 to the DMCNT.DTE bit. The DMSTS.ESIF flag is automatically cleared to 0 (interrupt source cleared), and DMA transfer is resumed.

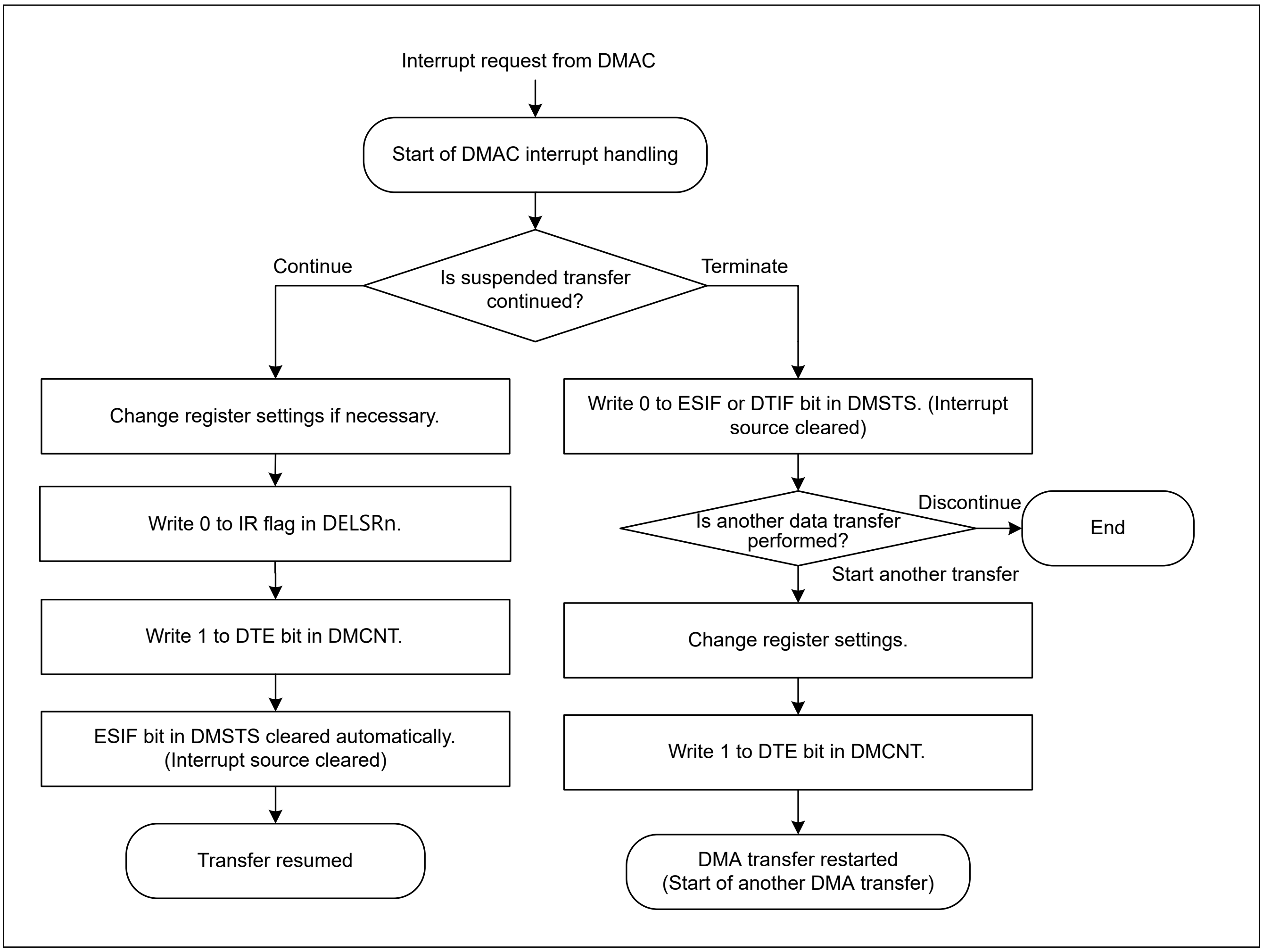

Figure 17.23 DMAC Interrupt Handling Routine to Resume/Terminate DMA Transfer

17.6.2 Transfer Error Interrupt

Error response detection interrupt request DMAm_TRANSERR (m = 0, 1) is generated from the DMAC/DTC when the transfer error is detected during DMAC transfer.

The types of interrupts that occur when a DMAC transfer error occurs are listed in the Table 17.20. The Table 17.20 also shows error information stored when a transfer error occurs.

Table 17.20 Interrupt and error information due to DMAC transfer error cause

Transfer error factor	NMI/RESET*1 Request	Interrupt Request	Bus Error Status	Error Channel Information
MSAU Error	ICU.NMISR.BUSST	DMAm_TRANSERR*1	BUS.BUSERRSTATDMACm.MESRRSTAT	DMAC. DMECHR
Master MPU Error	ICU.NMISR.BUSST	DMAm_TRANSERR*1	BUS.BUSERRSTATDMACm.MMERRSTAT	DMAC. DMECHR
Illegal Access Error	ICU.NMISR.BUSST	DMAm_TRANSERR*1	BUS.BUSERRSTATDMACm.ILERRSTAT	DMAC. DMECHR
Slave Bus Error	ICU.NMISR.BUSST	DMAm_TRANSERR*1	BUS.BUSERRSTATDMACm.SLERRSTAT	DMAC. DMECHR
TrustZone Filter error	ICU.NMISR.BUSST	DMAm_TRANSERR*1	BUS.BUSERRSTATDMACm.SLERRSTAT	DMAC. DMECHR

Note 1. If ICU.NMIER.BUSEN is enabled with DMAm_TRANSERR set in ICU.IELSR, NMI and an interrupt will occur due to a transfer error caused by DMA. Only NMI can be generated by not setting DMAm_TRANSERR to ICU.IELSR.

17.7 Event Link

Each DMAC channel outputs an event link request each time one data transfer is completed in normal transfer mode or repeat transfer mode, and each time one block transfer is completed in block transfer mode or repeat-block transfer mode.

The event link request DMACmn_INT (m = 0, 1, n = 0 to 7) is generated when the bus responds to the last destination access of DMA transfer.

The timing of the bus response to a DMA write access depends on the DMBWR.BWE and transfer destination. For details, see section 17.9.1. Bus Response to DMA Write Access.

For details, see section 19, Event Link Controller (ELC).

17.8 Low-Power Consumption Function

Before entering the module-stop state, Software Standby mode or Deep Software Standby mode, you must first set the DMAST.DMST bit to 0 (the DMAC module suspended) and use the settings in the sections that follow.

(1) Module-stop function

Setting the module-stop bit to 1 stops the DMAC corresponding to that module-stop bit.

If the module-stop bit is set to 1 while DMAC transfers data, the DMAC corresponding to that bit becomes the module-stop state after completing it.

When the module-stop bit is 1, access to the register of the DMAC corresponding to that module-stop bit is prohibited.

(2) Software Standby mode and Deep Software Standby mode

Use the settings described in section 11.6.2.1. Entering Software Standby Mode, or in section 11.6.3.1. Transition to Deep Software Standby Mode .

If DMA transfer operations are in progress when the WFI instruction is executed, the DMA transfer completes before the transition to Software Standby mode or Deep Software Standby mode.

(3) Notes on low power consumption function

For information on the WFI instruction and register settings, see section 11.8.7. Timing of WFI Instructions.

To perform a DMA transfer after returning from a low power mode, set the DMAST.DMST bit to 1 again. To use a request that is generated in Software Standby mode as an interrupt request to the CPU but not as a DMAC startup request, specify the CPU as the interrupt request destination, as described in section 14.4.1. Interrupt Detection Selection, and then execute the WFI instruction.

17.9 Usage Notes

17.9.1 Bus Response to DMA Write Access

The timing of the bus response to a DMA write access depends on the DMBWR.BWE setting and the destination of the transfer. When DMBWR.BWE is "1", the bus response is returned from the intermediate point described in section 15.2.5.2. Bufferable Write Access in the bus chapter. Also, when writing to an access destination described in section 15.2.5.3. Notes on Writing Completion Timing in the bus chapter, a bus response is returned before the write is completed.

17.9.2 Access to the Registers during DMA Transfer

Do not write to the following registers while the DMSTS.ACT flag of the same channel is set to 1 (DMAC active state) or the DMCNT.DTE bit of the same channel is set to 1 (DMA transfer enabled):

- ICU.DELSR
- DMSAR
- DMDAR
- DMCRA
- DMCRB
- DMTMD
- DMINT
- DMAMD
- DMOFR

- DMSBS
- DMDBS
- DMSRR
- DMDRR
- DMBWR
- ICUSARC
- DMACSAR
- DMCTL

17.9.3 DMA Transfer to Reserved Areas

DMA transfer to the reserved areas is prohibited. If such an access is made, transfer results are not guaranteed. For details on the reserved areas, see section 5, Address Space.

17.9.4 Setting of DMAC Event Link Setting Register of the Interrupt Controller Unit (ICU.DELSRn)

The DMAC event link setting register (ICU.DELSRn) should be set while the DMA transfer enable bit (DMCNT.DTE) is cleared to 0 (DMA transfer is disabled). Moreover, the DTC activation enable register (ICU.IELSRn.DTCE) that corresponds to the same event number that has been set by the ICU.DELSRn register should not be set to 1. For details on the ICU.IELSRn.DTCE, see section 14, Interrupt Controller Unit (ICU).

17.9.5 Suspending or Restarting DMAC Activation

To suspend a DMAC activation request, write 0x000 to the DMAC Event Link select bits (ICU.DELSRn.DELS[9:0]). To restart the DMA transfer, write the event number to the ICU.DELSRn.DELS[9:0] bits following the settings shown in section 17.3.9. Activating the DMAC.

17.9.6 Precautions for Resuming DMA Transfer

A DMAC activation request might occur in the next request after a DMA transfer completes. If this happens, the DMA transfer starts and the DMAC activation request is held in the DMAC. To prevent this, stop the DMAC activation requests by setting the ICU.DELSRn.DELS[9:0] bits to 0.

When a DMAC activation request occurs after the last round of the DMA transfer is generated, clear the DMAC activation request with the following steps:

1. Set the DMCNT.DTE bit to 0.
2. Set the ICU.DELSRn.IR flag to 0.